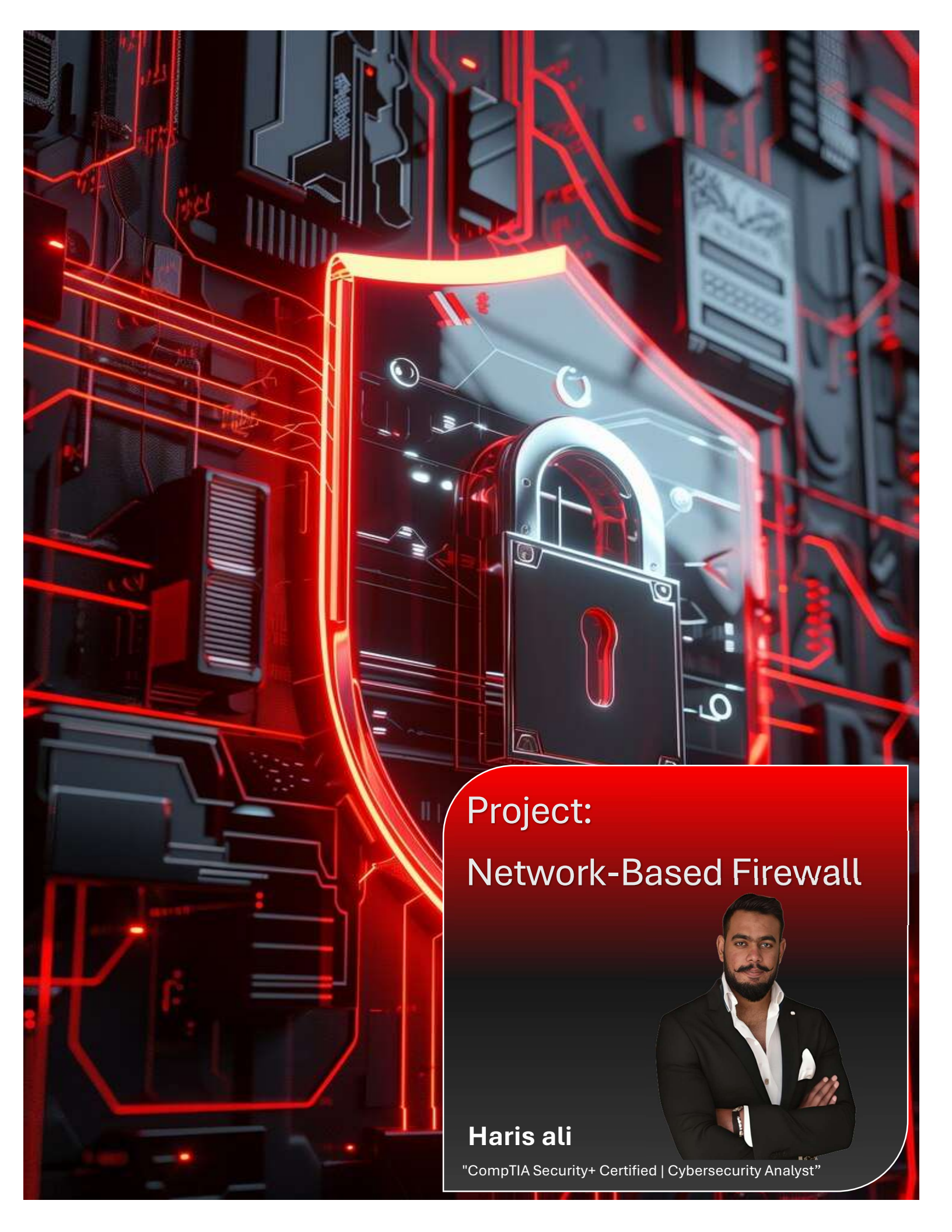




Project:

Network-Based Firewall



Haris ali

"CompTIA Security+ Certified | Cybersecurity Analyst"

Exercise:

Implement Network-Based Firewall Functionality: Block Insecure Ports using pfSense Firewall

Firewall rules allow a computer to send or receive packets from a program, services, computers, and/or users.

Lab Scenario

To keep the computer resources of the organization secure, the security professional needs to configure outbound traffic because outbound traffic leaves the network vulnerable to malware that targets organizational resources. These threats can be protected by using firewall rules. The pfSense firewall allows specific traffic on specific ports while blocking all other traffic.

Lab Objectives

This lab will demonstrate how to block insecure ports using the pfSense firewall and protect endpoints within the network using the pfSense firewall.

Overview of Firewall Rules

Firewall rules can be created for either inbound or outbound traffic.

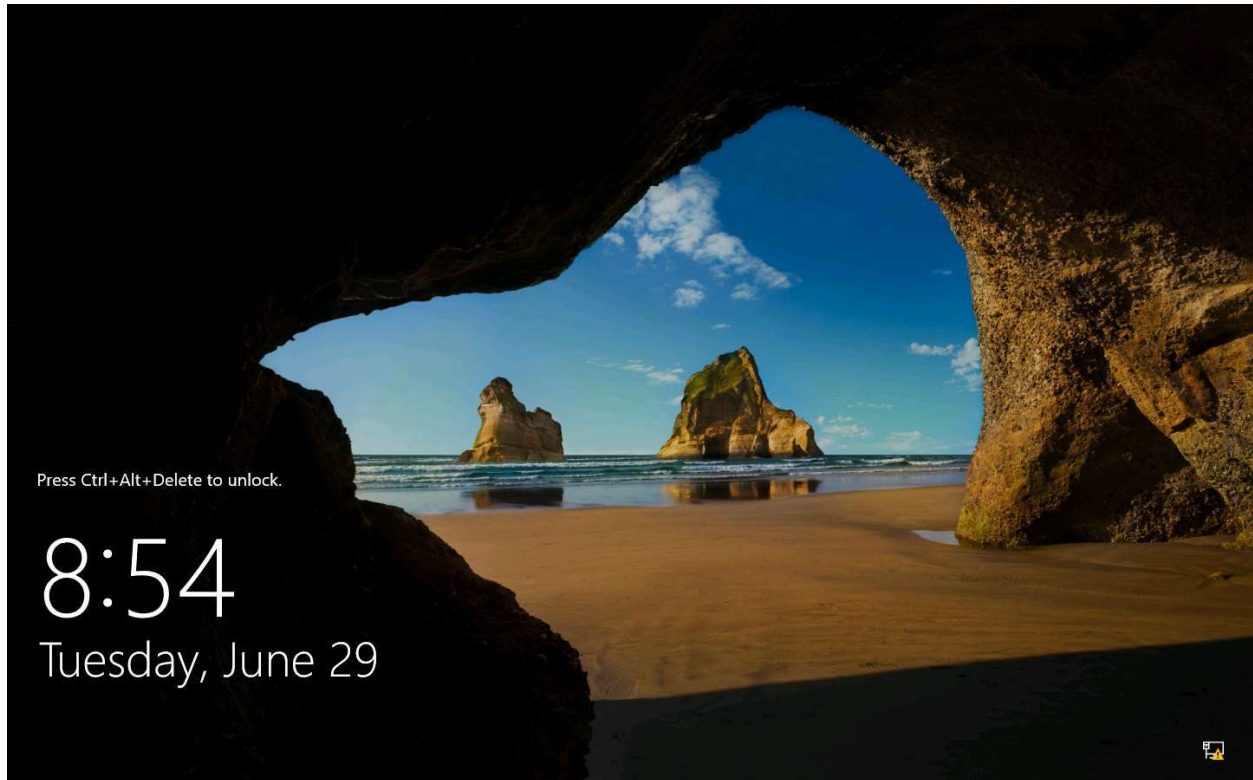
- An inbound firewall rule protects the network against incoming malicious traffic from the Internet or other network segments.
- An outbound firewall protects against outgoing traffic originating inside an enterprise network.

Firewall rules can be configured to specify computers, users, programs, services, ports, and protocols.

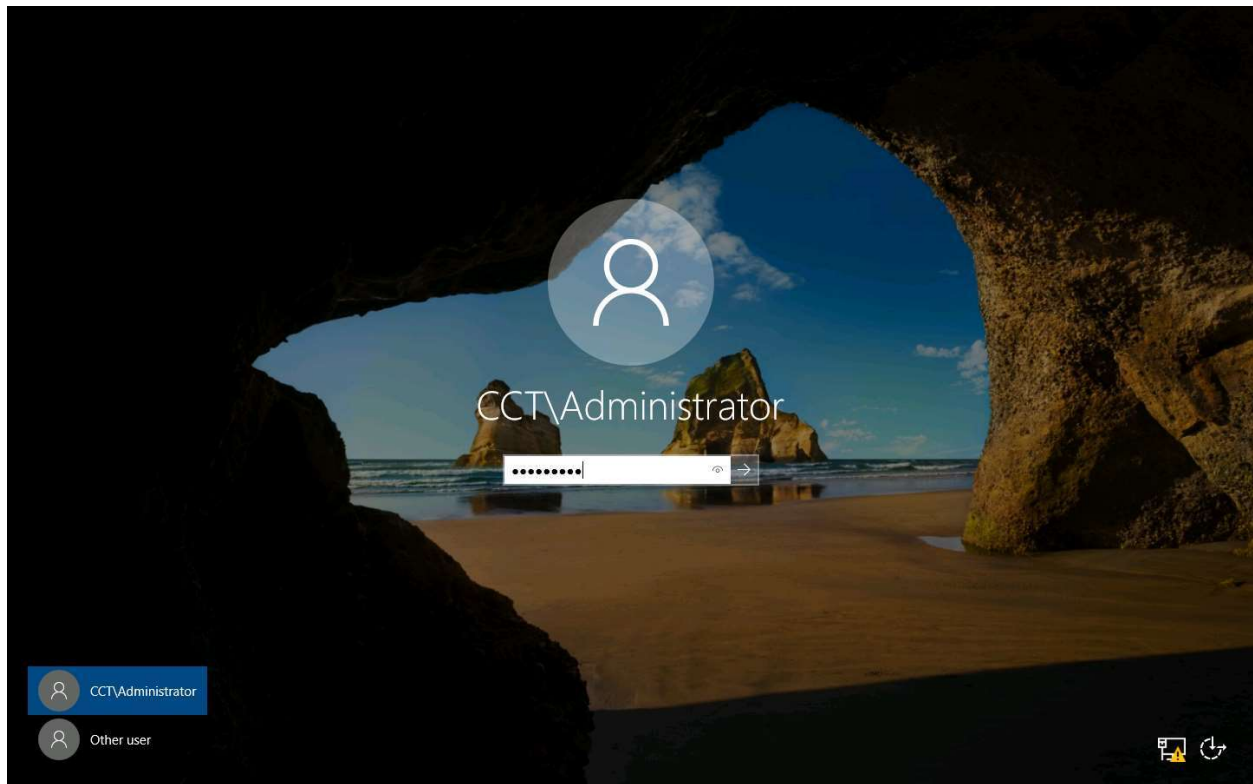
Lab Tasks

If you have already launched AD Domain Controller, and Web Server machines in the previous exercise, skip to step#5.

1. Click [AD Domain Controller](#) to launch the AD Domain Controller machine. Click [Ctrl+Alt+Delete](#) link to login to AD Domain Controller.

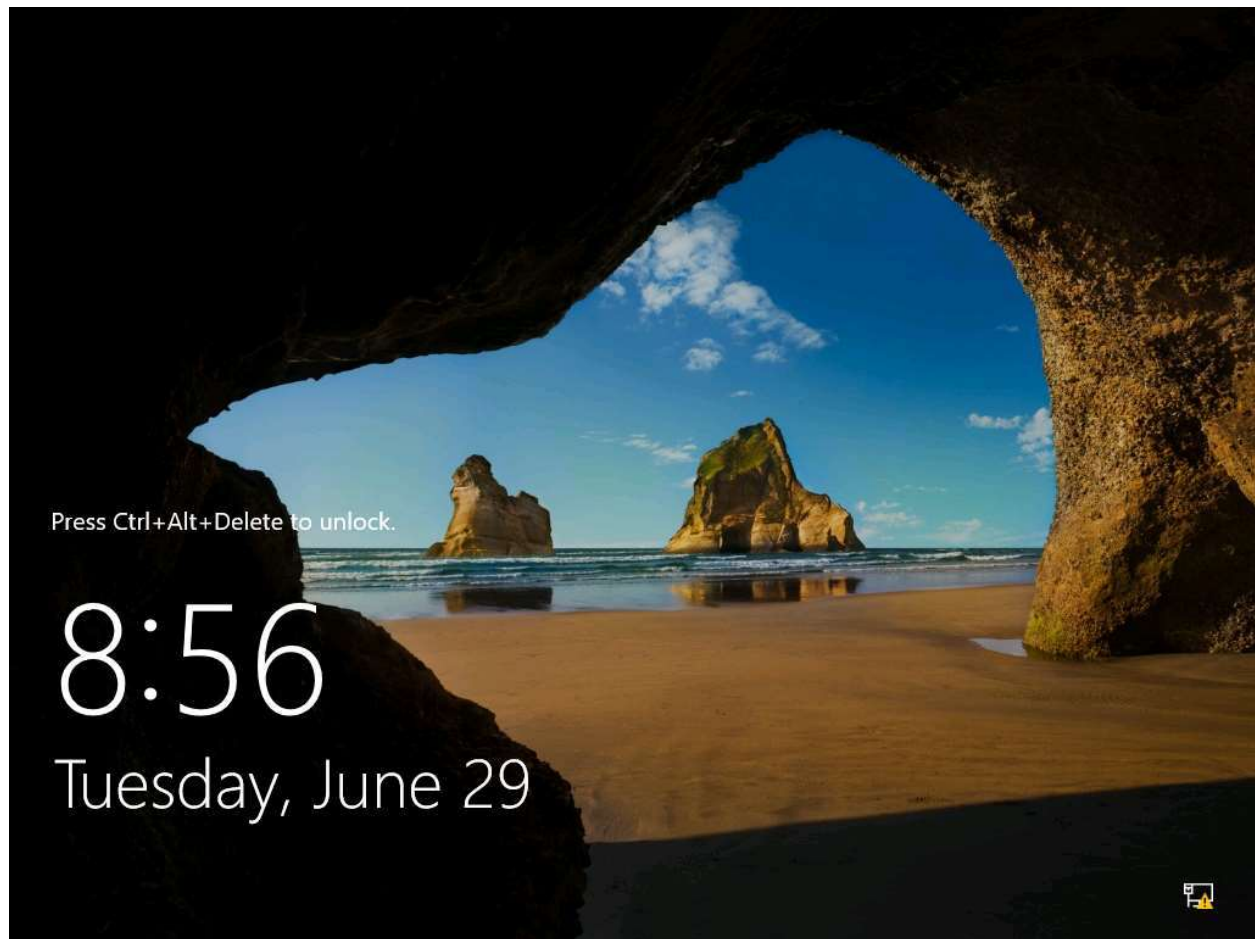


2. By default, the CCT\Administrator account is selected. Click admin@123 and press Enter to login.

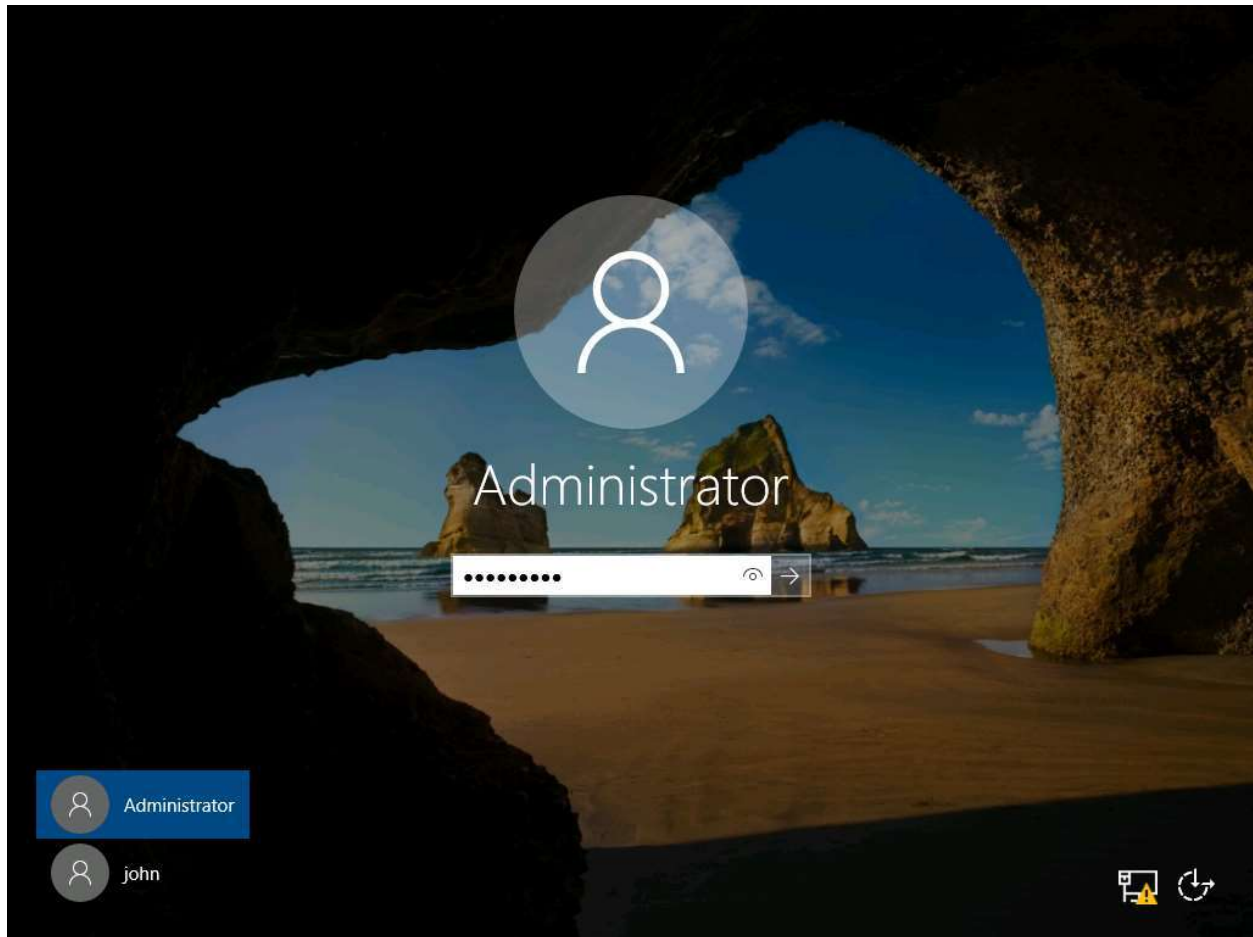


If the network screen appears, click Yes.

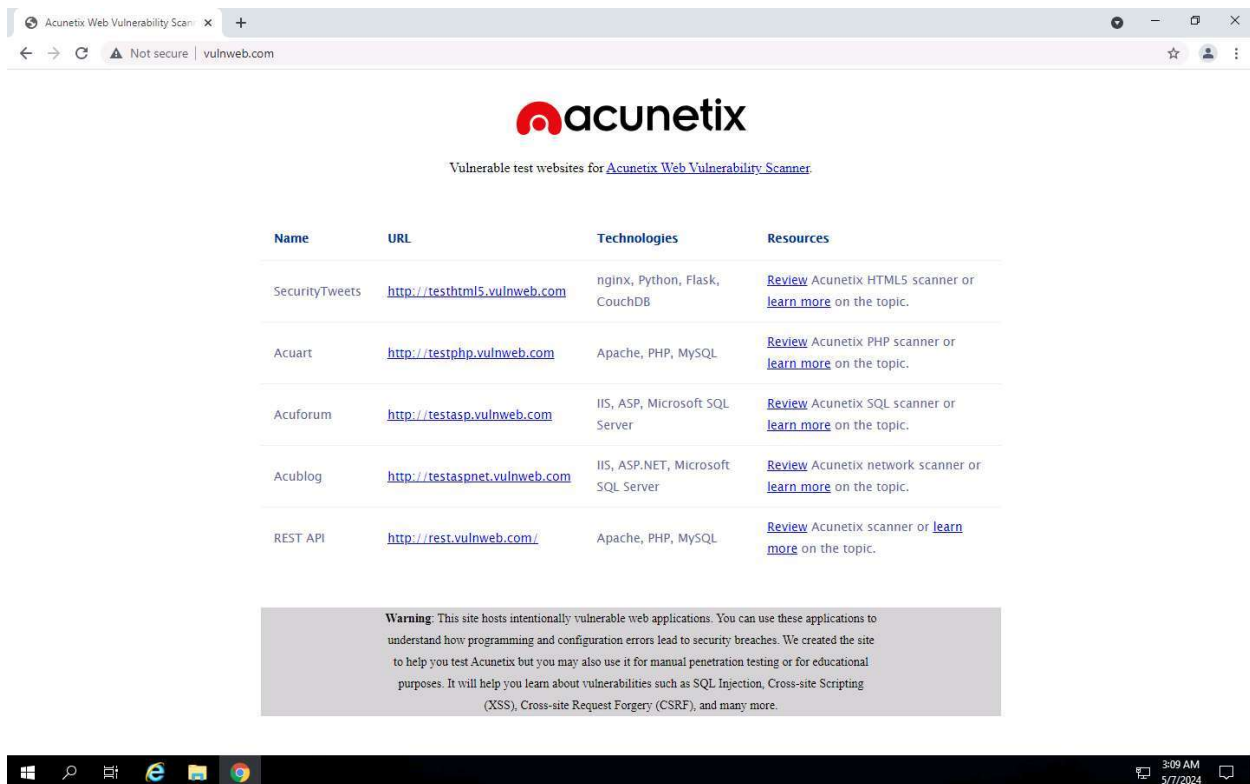
3. Click [Web Server](#) to launch the Web Server machine.
Click [Ctrl+Alt+Delete](#).



4. By default, the Administrator account is selected, click admin@123 and press Enter to login.



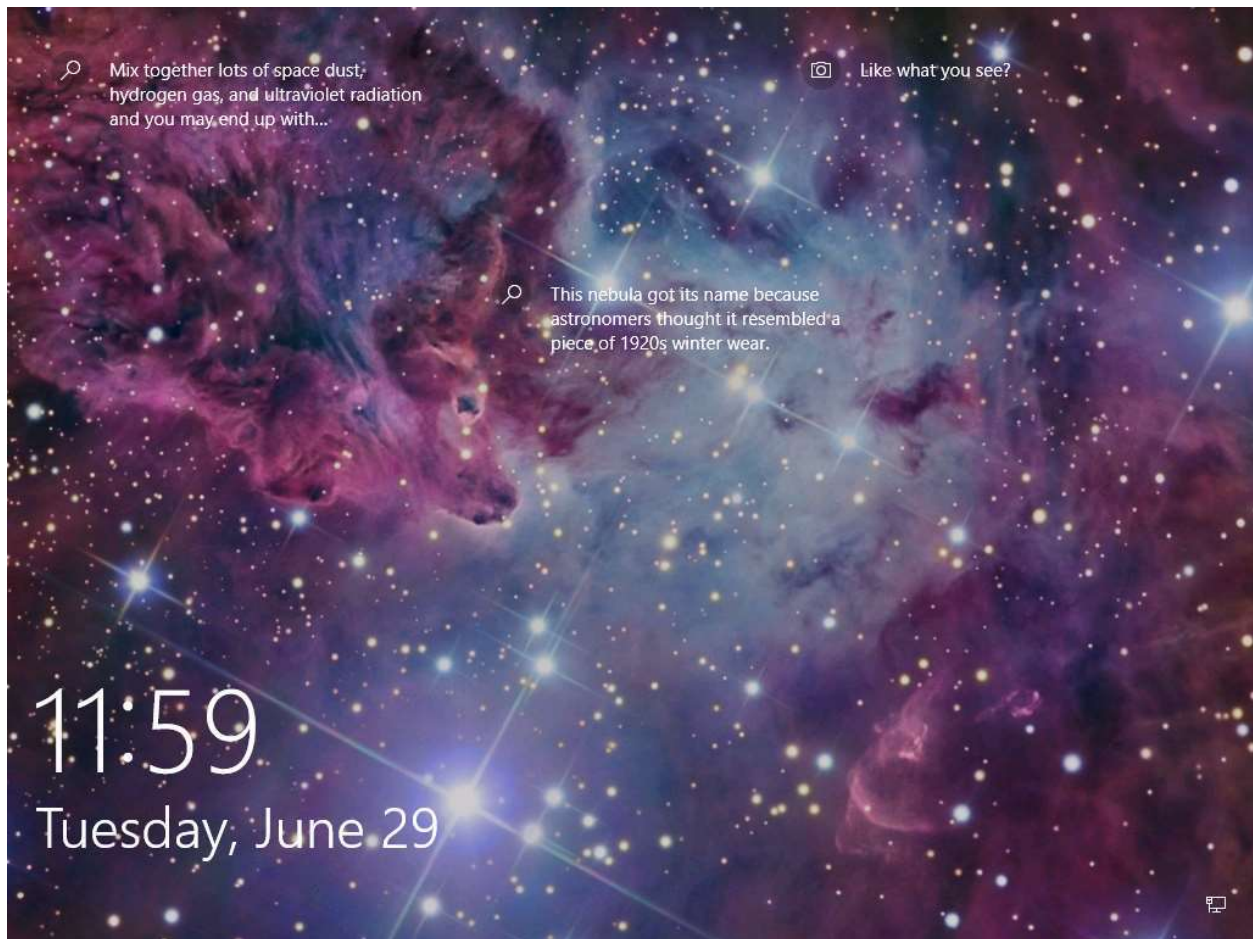
5. In Web Server machine, to open the browser, double click the Google Chrome icon on the Desktop.
6. Type `http://www.vulnweb.com` in the address bar, and press Enter. You will be able to access the web page, as shown in the below screenshot.



7. Next, we shall create a rule to restrict a user from accessing HTTP-enabled websites (by blocking port http 80), so that they can access only https-enabled websites on the Internet.

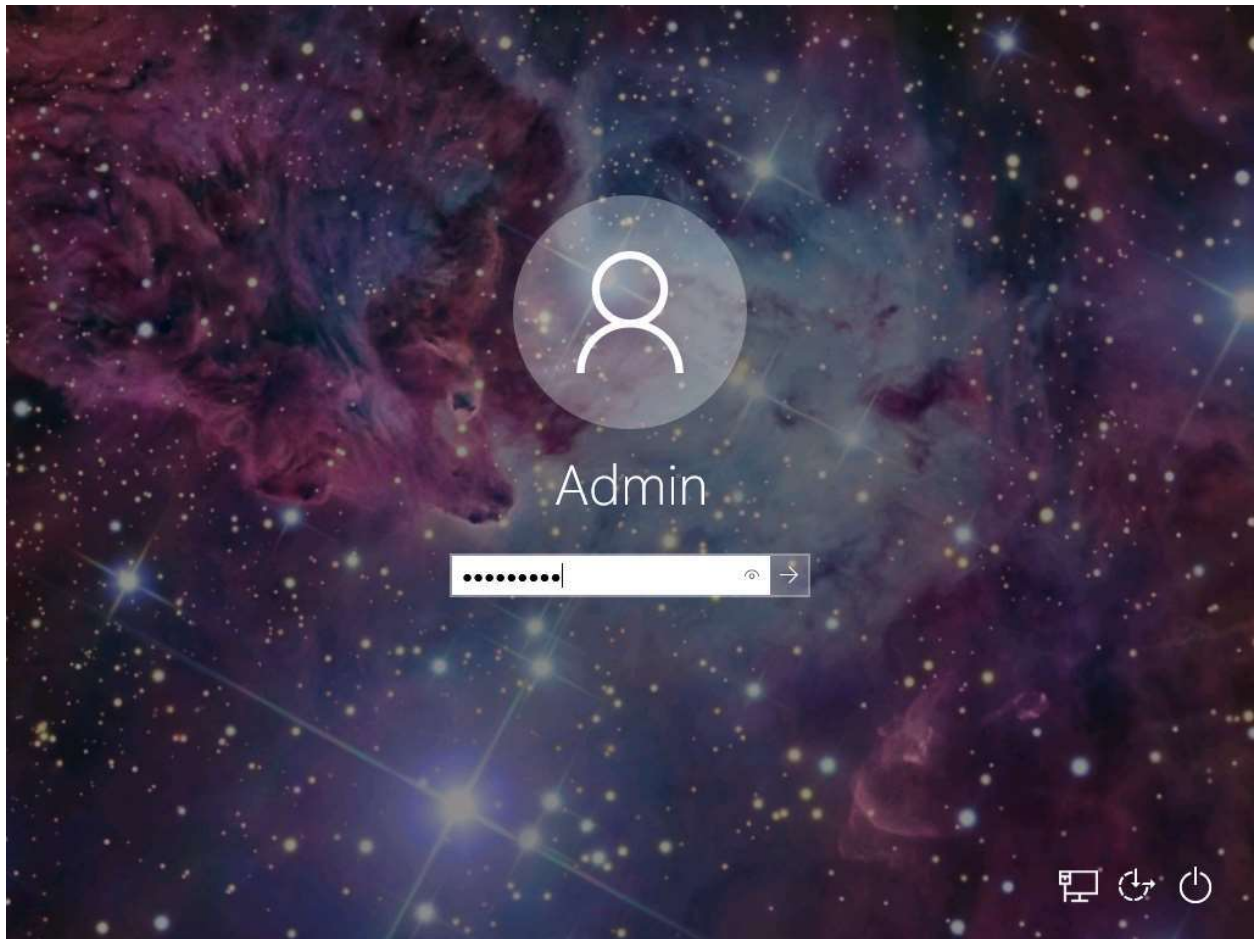
8. Click [Admin Machine-1](#) to launch the Admin Machine-1 machine. Click [Ctrl+Alt+Delete](#) link to login to Admin Machine-1.

If already logged in to Admin Machine-1, skip step 9.

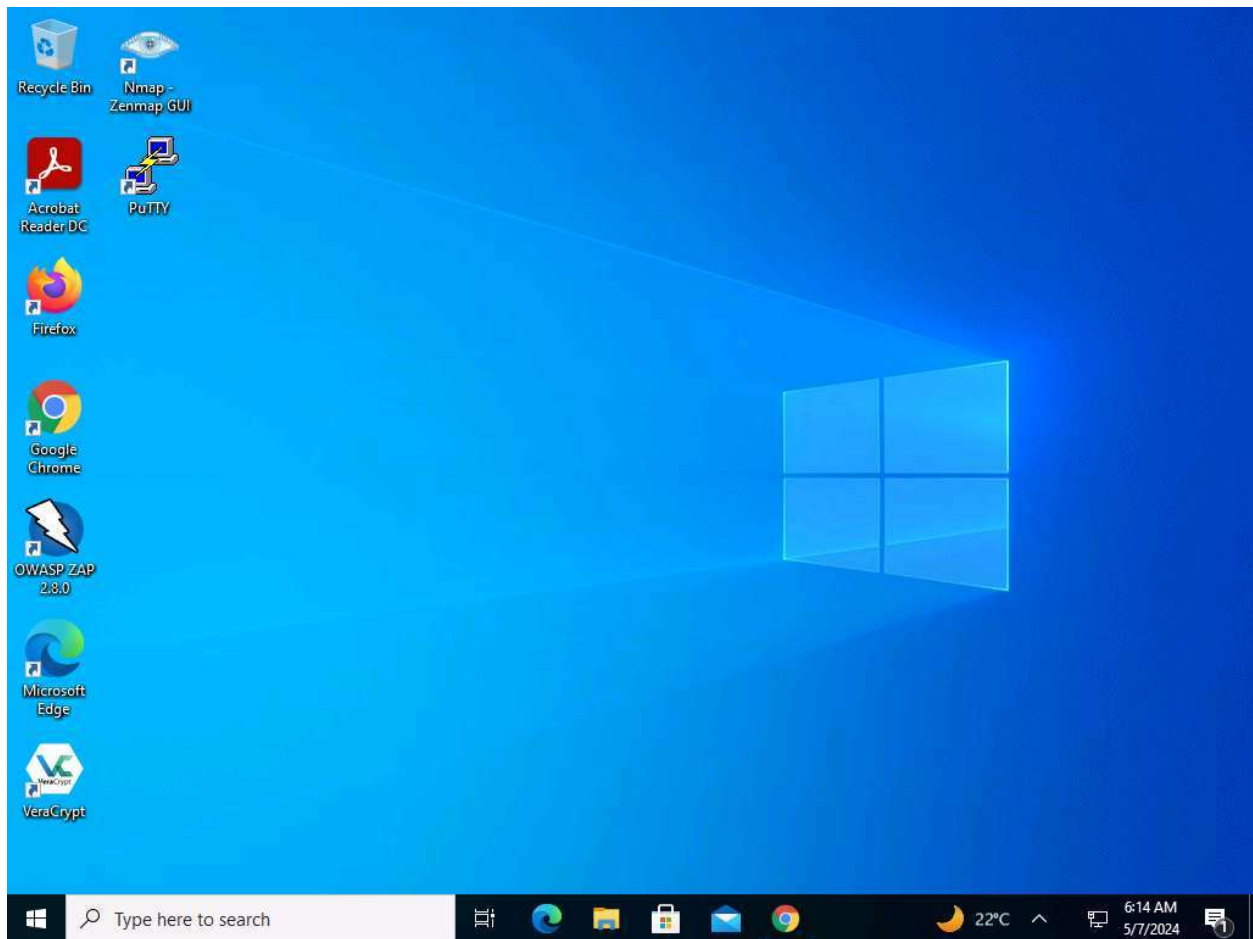


9. By default, the Admin account is selected. Click `admin@123` and press Enter to login.

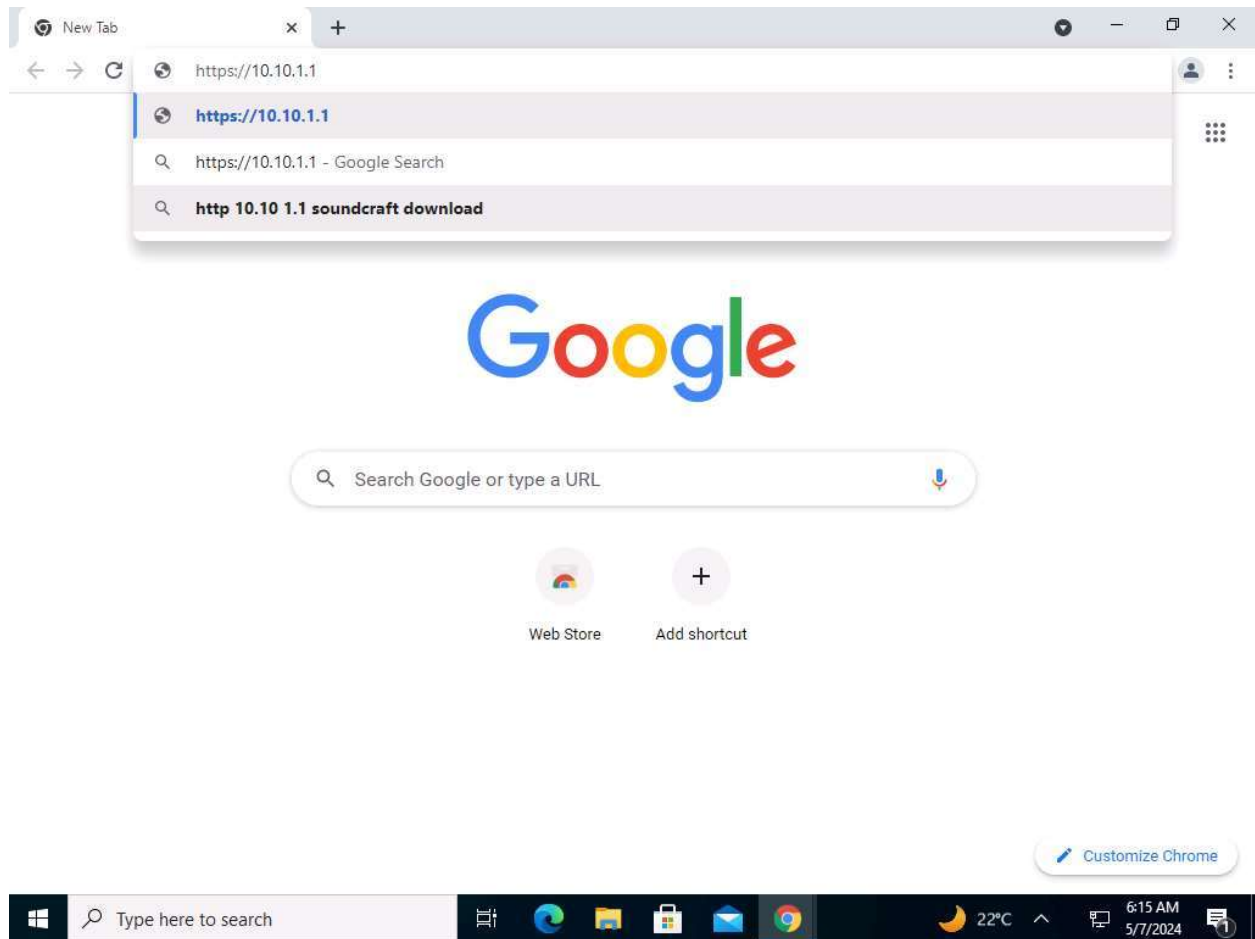
If the network screen appears, click Yes.



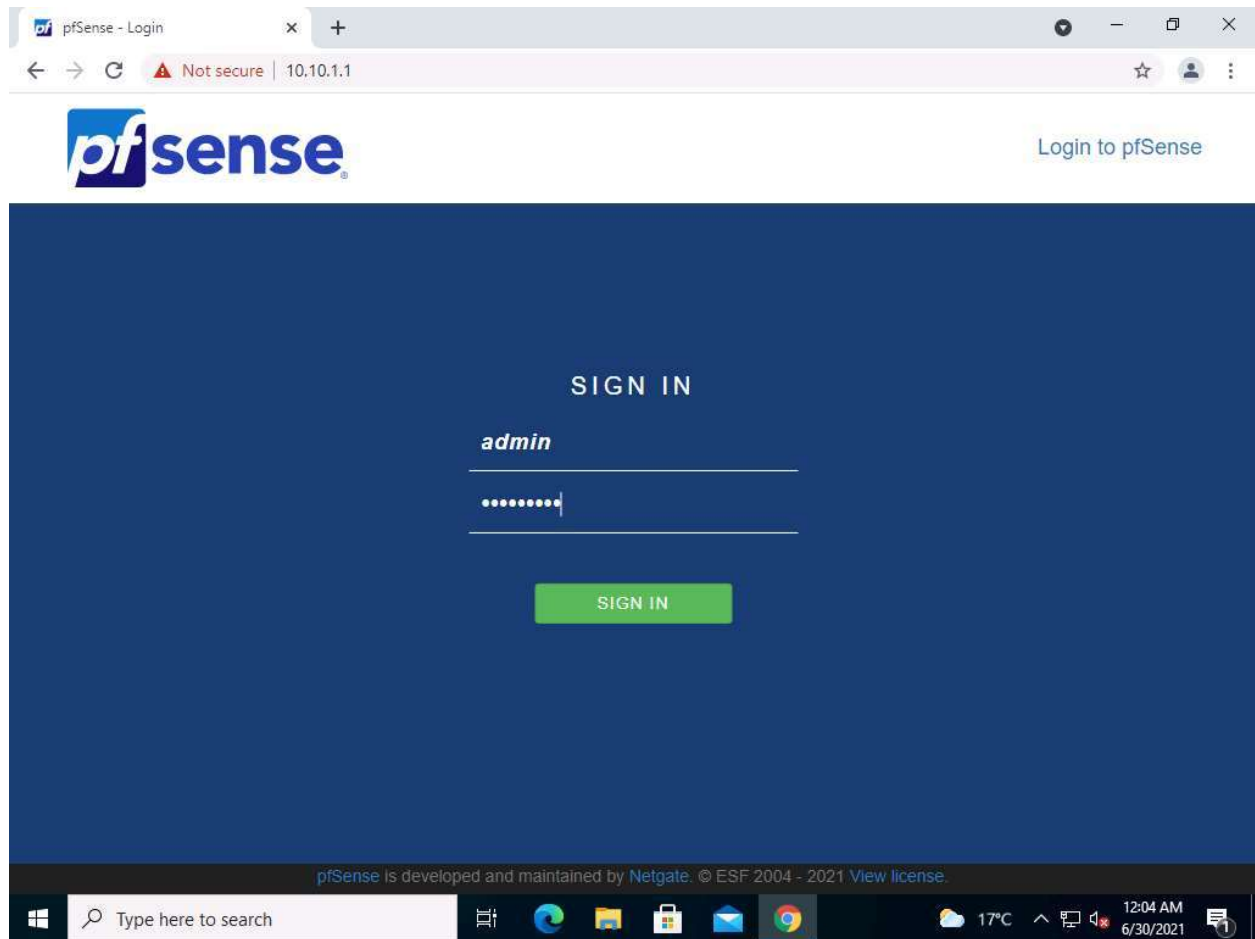
10. To open the browser, double click the Google Chrome icon on the Desktop.



11. Browse pfSense web interface. Type <https://10.10.1.1> in the address bar, and press Enter. Click Advanced button and click proceed to 10.10.1.1 (unsafe) link.



12. The pfSense login page will appear. Type the username as admin and password as admin@123, and click the SIGN IN button, as shown in the screenshot below.



13. The pfSense Dashboard will appear. Navigate to Firewall-->Rules from the main menu.

The screenshot shows the pfSense web interface in a browser window. The browser's address bar displays 'pfSense.localdomain - Status: D...' and the URL 'https://10.10.1.1/firewall_rules.php'. The pfSense header includes the logo and navigation tabs: System, Interfaces, Firewall (selected), Services, VPN, Status, Diagnostics, and Help. A dropdown menu for the Firewall tab is open, showing options: Aliases, NAT, Rules (highlighted with a red box), Schedules, Traffic Shaper, and Virtual IPs. The main content area is divided into two panels. The left panel, titled 'System Information', contains a table with details about the pfSense instance, including its name, user, system information, BIOS details, version, CPU type, kernel PTI, and MDS Mitigation status. The right panel, titled 'Netgate Services And Support', provides information about the support contract type and links to various support resources.

System Information	
Name	pfSense.localdomain
User	admin@10.10.1.2 (Local D...
System	Microsoft Azure Netgate Device ID: 9419252c0bb6c3ee4f14
BIOS	Vendor: American Megatrends Inc. Version: 090006 Release Date: Thu Apr 28 2016
Version	2.4.5-RELEASE-p1 (amd64) built on Tue Jun 02 17:51:17 EDT 2020 FreeBSD 11.3-STABLE Version 2.5.1 is available. Cloud icon Version information updated at Wed Jun 30 3:37:41 UTC 2021 Refresh icon
CPU Type	Intel(R) Xeon(R) Gold 6262V CPU @ 1.90GHz AES-NI CPU Crypto: Yes (inactive)
Kernel PTI	Disabled
MDS Mitigation	Inactive

Netgate Services And Support

Contract type: Community Support
Community Support Only

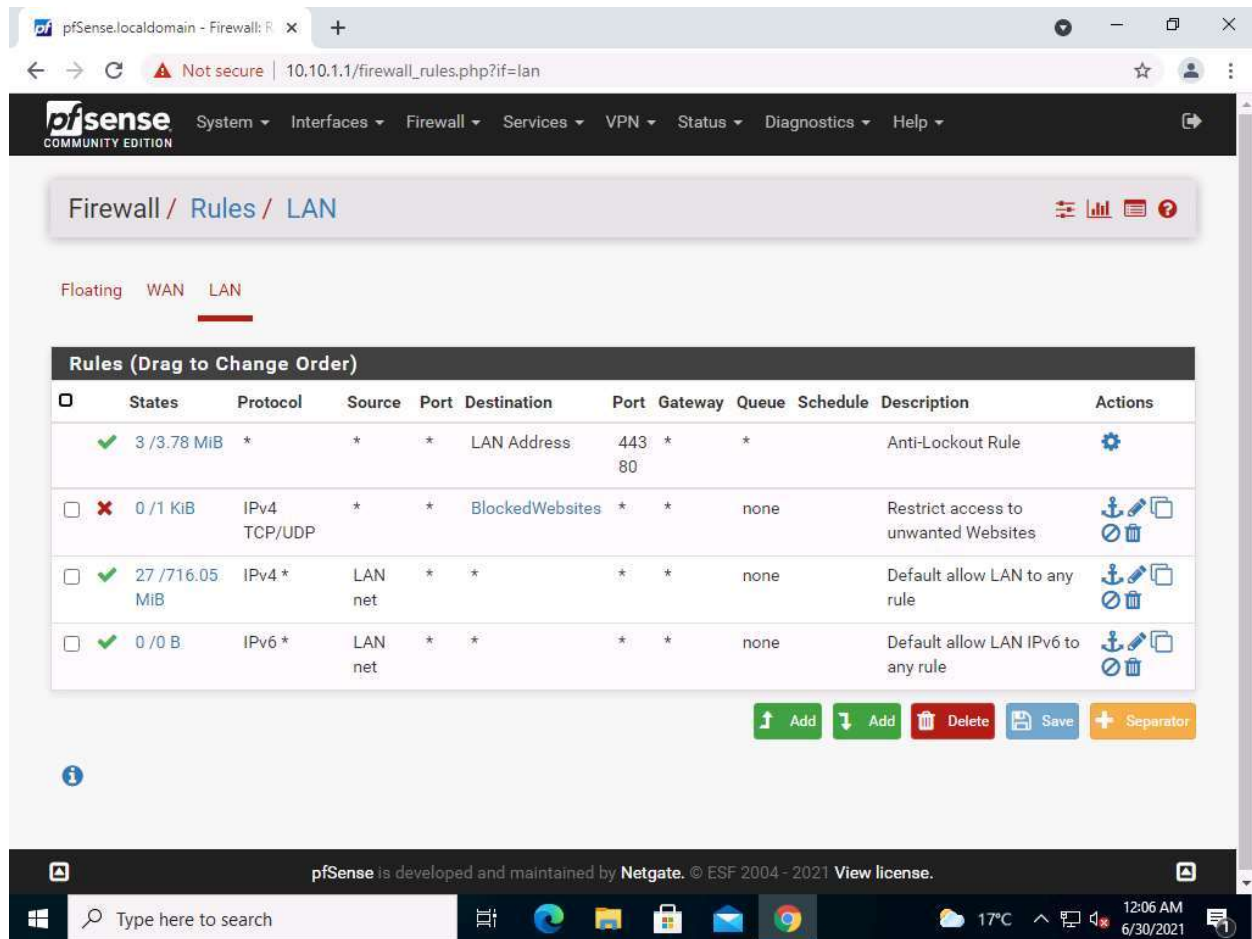
NETGATE AND pfSense COMMUNITY SUPPORT RESOURCES

If you purchased your pfSense gateway firewall appliance from Netgate and elected **Community Support** at the point of sale or installed pfSense on your own hardware, you have access to various community support resources. This includes the **NETGATE RESOURCE LIBRARY**.

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- [Netgate Global Support FAQ](#)
- [Official pfSense Training by Netgate](#)
- [Netgate Professional](#)
- [Visit Netgate.com](#)

14. The Firewall/Rules/WAN page will appear. Click the LAN option.



15. To create a rule, click the up arrow Add button.

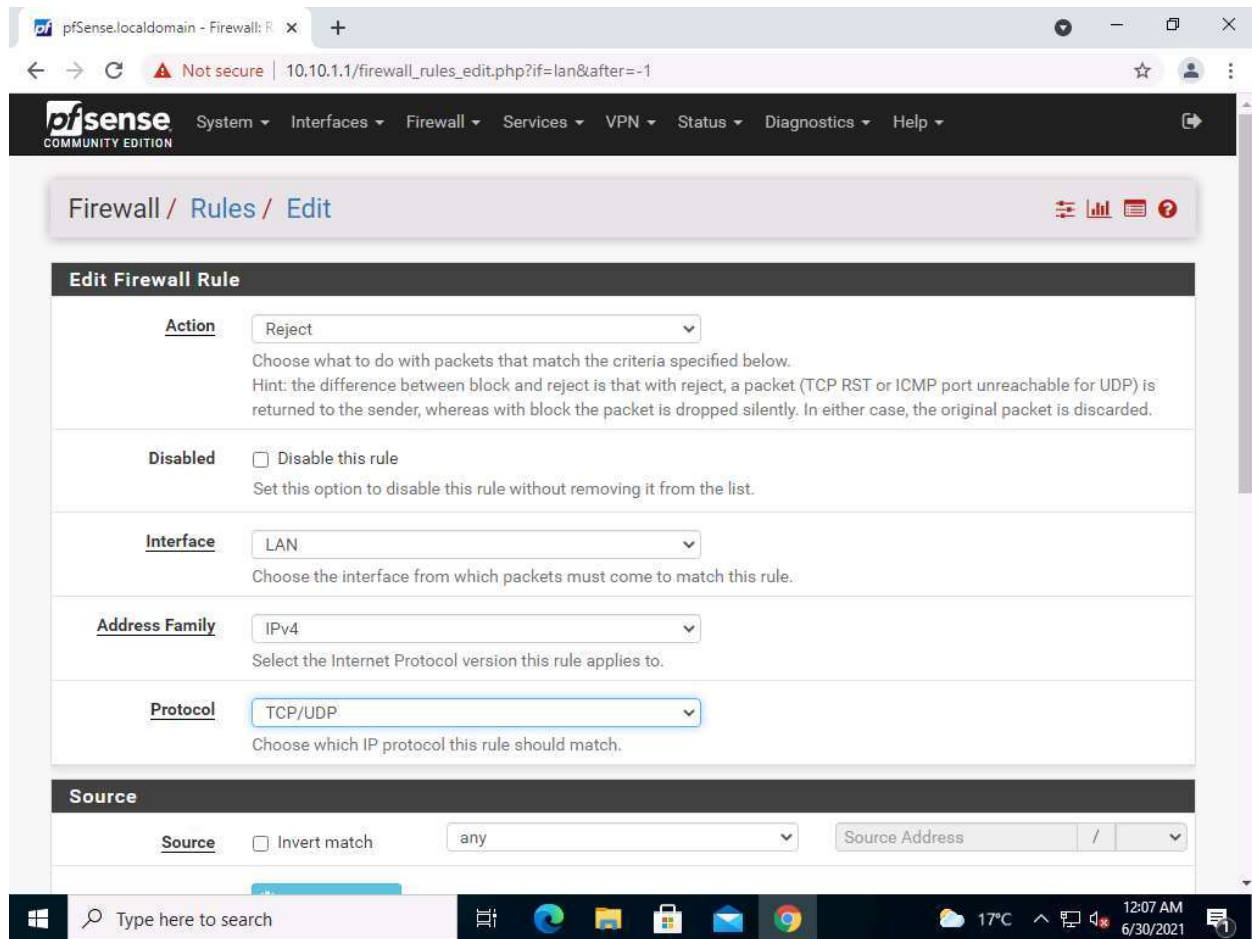
16. Set the following details under Edit Firewall Rule section:

Action --> Reject

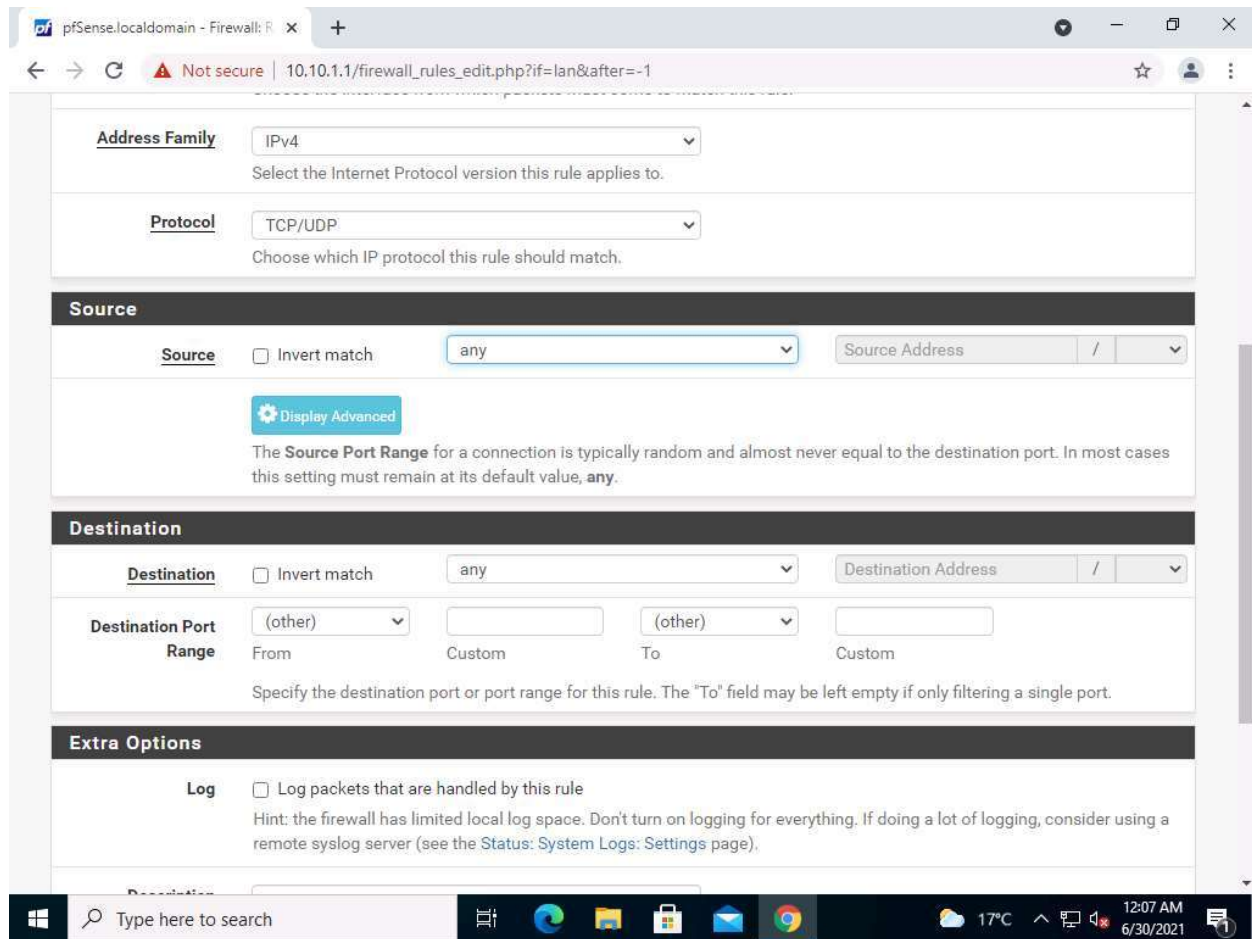
Interface --> LAN

Address Family --> IPv4

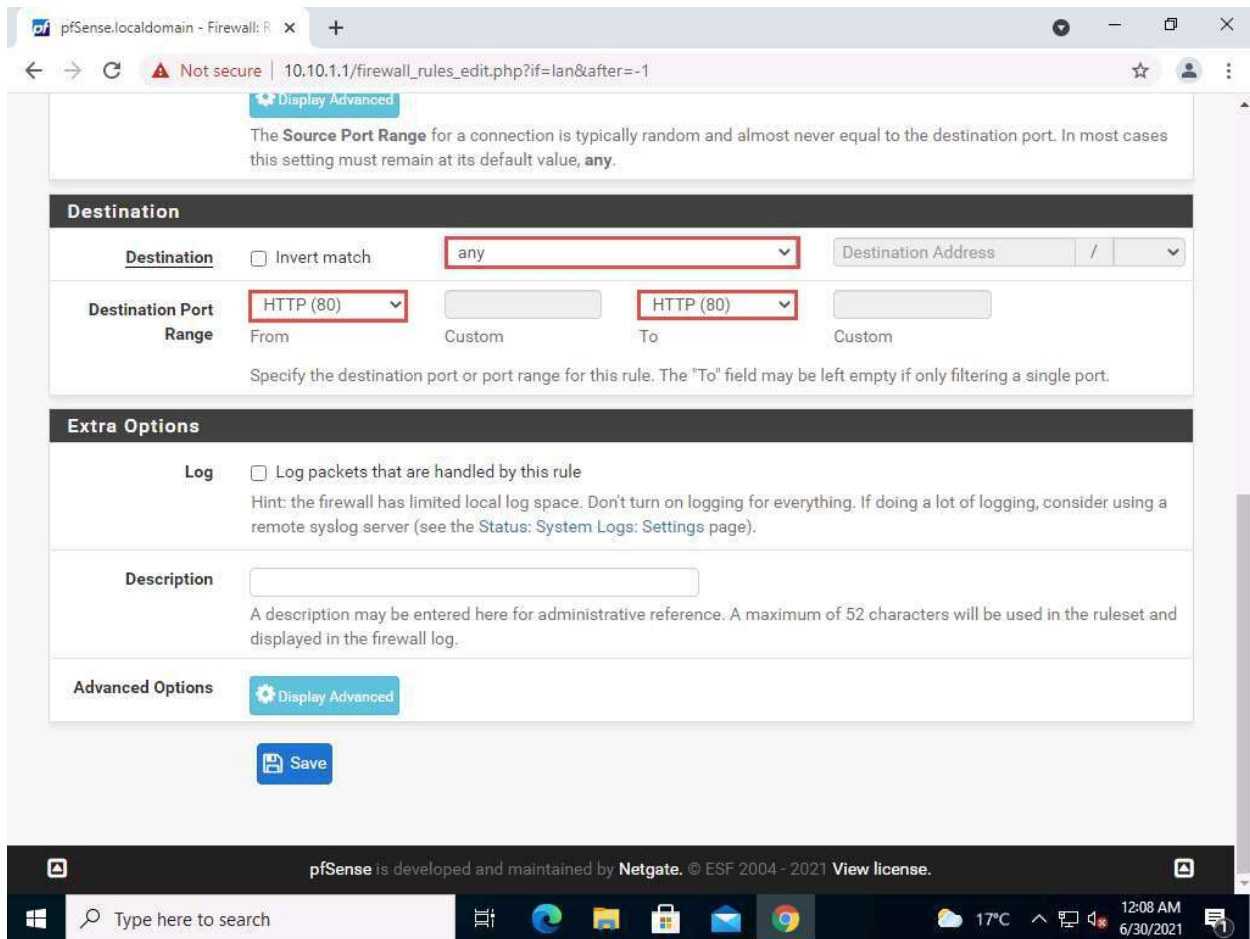
Protocol --> TCP/UDP.



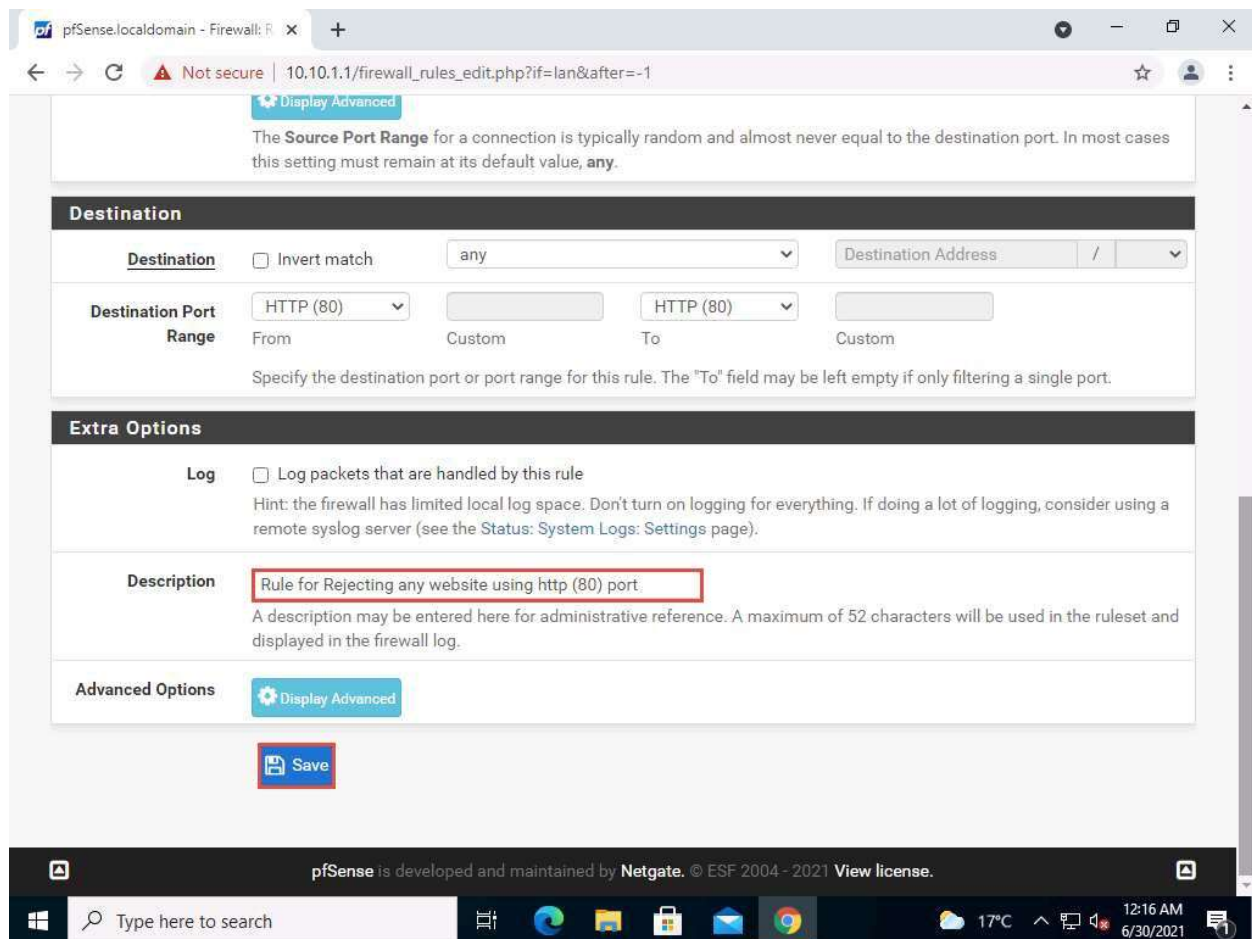
17. Under Source section, select any from the dropdown as shown in the below screenshot.



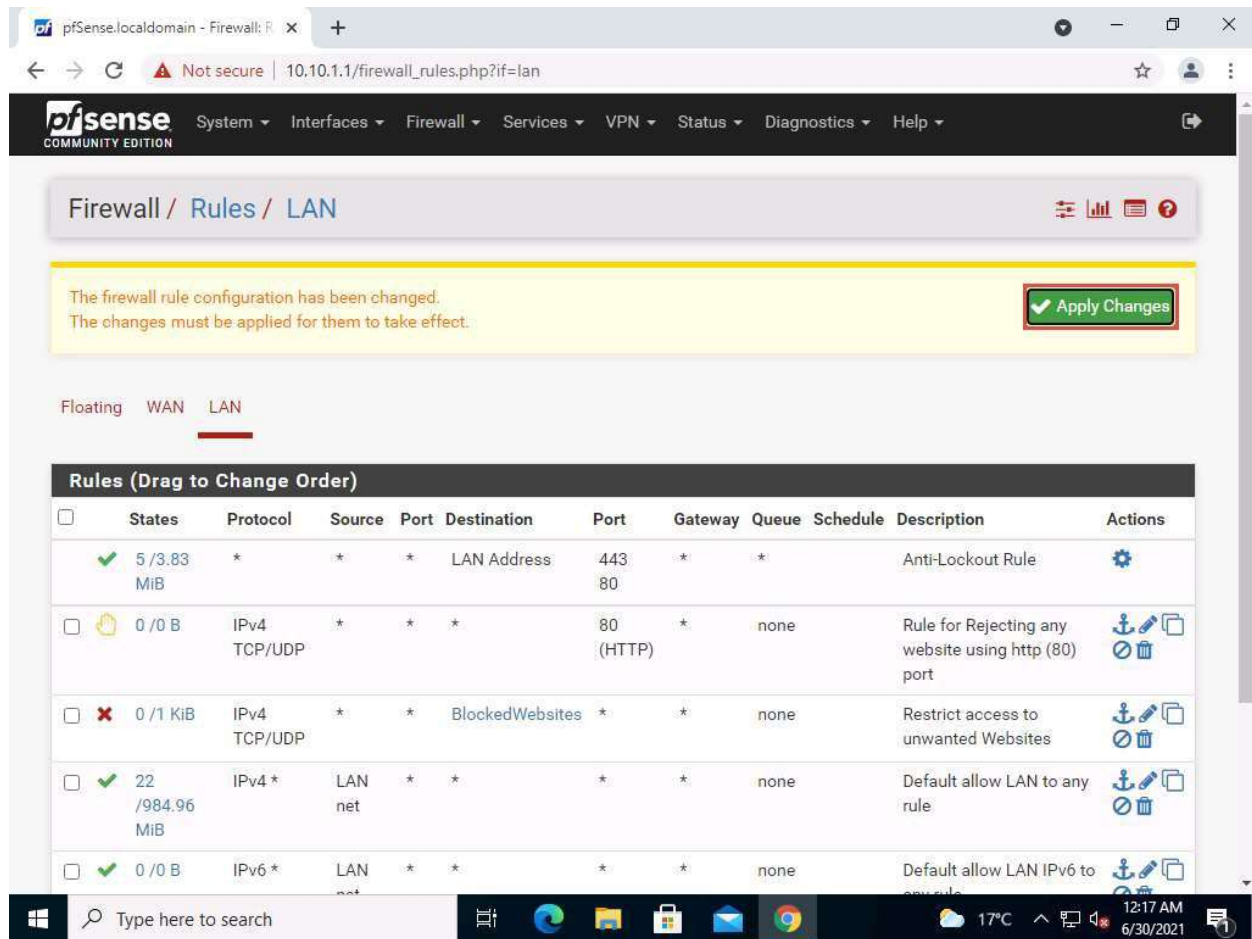
18. Under Destination section, select any from the dropdown and set Destination Port Range to HTTP (80) from the dropdown.



19. Scroll down. Under Extra Options, enter Rule for Rejecting any website using http (80) port in the Description field, and click Save.

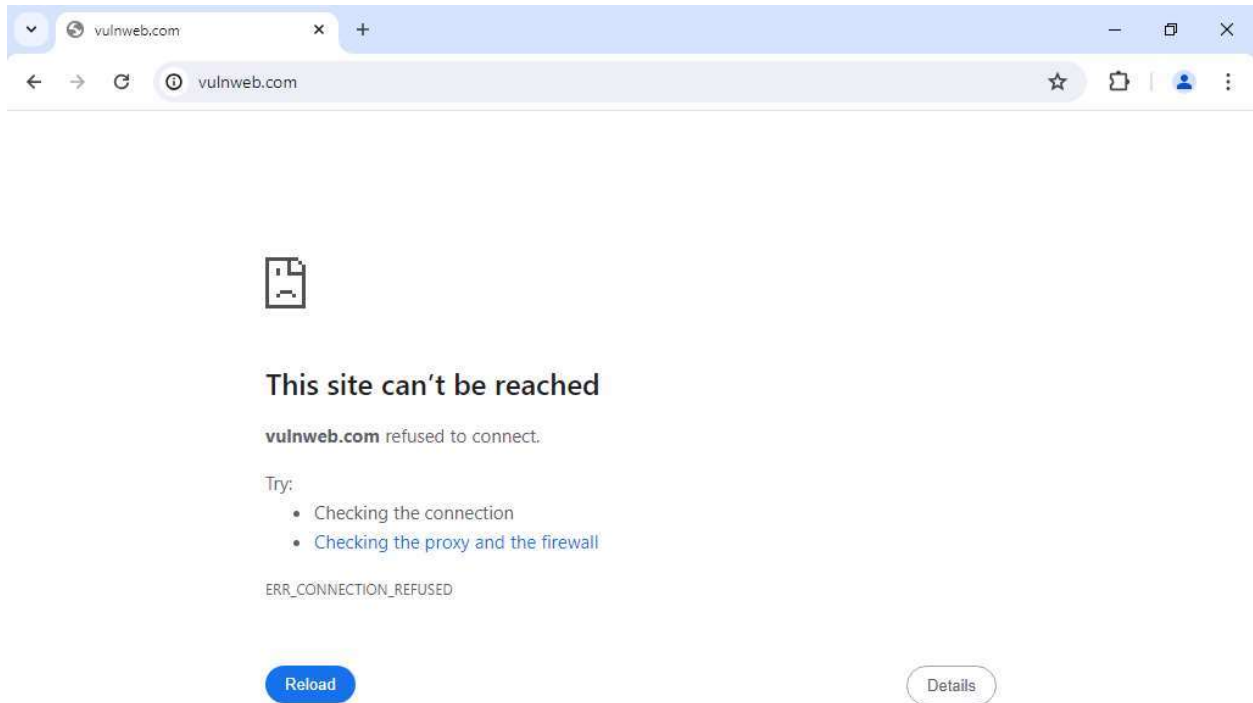


20. The page will redirect to the Firewall/Rules/LAN page. Click Apply Changes.



21. The firewall rule has been successfully created.
22. Close browser window.
23. Click [Web Server](#) to switch back to the Web Server machine.
24. In the Chrome browser (it is already opened or you can launch a new window), enter `http://vulnweb.com/` in the address bar, and press Enter. You will see the message as This site can't be reached. This is because the pfSense firewall rule is now preventing the traffic from the port http.

If changes are not affecting, then reboot the pfSense firewall and clear the Chrome browser cache.



25. Close browser window.
26. Delete the firewall rules created in this exercise and previous exercise.
27. To delete the firewall rules, click [Admin Machine-1](#) to switch back to the Admin Machine-1 machine. Repeat steps Step 10 to Step 12. Navigate to Firewall->Rules from the main menu.

The screenshot shows the pfSense Community Edition web interface. The 'Firewall' menu is open, and the 'Rules' option is highlighted with a red box. The background shows the 'Status / Dashboard' page with system information and support resources.

System Information

Name	pfSense.localdomain
User	admin@10.10.1.2 (Local D
System	Microsoft Azure Netgate Device ID: 9419252c0bb6c3ee4f14
BIOS	Vendor: American Megatrends Inc. Version: 090006 Release Date: Thu Apr 28 2016
Version	2.4.5-RELEASE-p1 (amd64) built on Tue Jun 02 17:51:17 EDT 2020 FreeBSD 11.3-STABLE Version 2.5.1 is available. Cloud icon Version information updated at Wed Jun 30 3:37:41 UTC 2021 Refresh icon
CPU Type	Intel(R) Xeon(R) Gold 6262V CPU @ 1.90GHz AES-NI CPU Crypto: Yes (inactive)
Kernel PTI	Disabled
MDS Mitigation	Inactive

Netgate Services And Support

Contract type: Community Support
Community Support Only

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- [Netgate Global Support FAQ](#)
- [Official pfSense Training by Netgate](#)
- [Netgate Professional](#)
- [Visit Netgate.com](#)

28. The Firewall/Rules/WAN page will appear. Click the LAN option. Select both the created rules by checking the checkboxes against the rules. and click Delete button.

pfSense.localdomain - Firewall: R x

Not secure | 10.10.1.1/firewall_rules.php?if=lan

System Interfaces Firewall Services VPN Status Diagnostics Help

Firewall / Rules / LAN

Floating WAN LAN

Rules (Drag to Change Order)

☐	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☐	✓ 1 / 6.54 MiB	*	*	*	LAN Address	443 80	*	*		Anti-Lockout Rule	
☒	0 / 3 KiB	IPv4 TCP/UDP	*	*	*	80 (HTTP)	*	none		Rule for Rejecting any website using http (80) port	
☒	✗ 0 / 2 KiB	IPv4 TCP/UDP	*	*	BlockedWebsites	*	*	none		Restrict access to unwanted Websites	
☐	✓ 20 / 986.31 MiB	IPv4 *	LAN net	*	*	*	*	none		Default allow LAN to any rule	
☐	✓ 0 / 0 B	IPv6 *	LAN net	*	*	*	*	none		Default allow LAN IPv6 to any rule	

Add
 Add
 Delete
 Save
 Separator

Type here to search

17°C 12:29 AM 6/30/2021

29. Click OK in the Rule deletion confirmation pop-up.

pfSense.localdomain - Firewall: R x

Not secure | 10.10.1.1/firewall_rules.php?if=lan

10.10.1.1 says
Are you sure you wish to delete selected rules?

OK Cancel

Firewall / Rules / LAN

Floating WAN LAN

Rules (Drag to Change Order)

☐	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☐	✓ 1 / 6.54 MiB	*	*	*	LAN Address	443 80	*	*		Anti-Lockout Rule	⚙️
☒	🚫 0 / 3 KiB	IPv4 TCP/UDP	*	*	*	80 (HTTP)	*	none		Rule for Rejecting any website using http (80) port	🔗 ⚙️ 🗑️
☒	🚫 0 / 2 KiB	IPv4 TCP/UDP	*	*	BlockedWebsites	*	*	none		Restrict access to unwanted Websites	🔗 ⚙️ 🗑️
☐	✓ 20 / 986.31 MiB	IPv4 *	LAN net	*	*	*	*	none		Default allow LAN to any rule	🔗 ⚙️ 🗑️
☐	✓ 0 / 0 B	IPv6 *	LAN net	*	*	*	*	none		Default allow LAN IPv6 to any rule	🔗 ⚙️ 🗑️

⬆️ Add ⬇️ Add 🗑️ Delete 💾 Save ➕ Separator

Type here to search

17°C 12:30 AM 6/30/2021

30. The page will redirect to the Firewall/Rules/LAN page. Click Apply Changes.

pfSense.localdomain - Firewall: R x +

Not secure | 10.10.1.1/firewall_rules.php?if=lan

pfSense COMMUNITY EDITION System Interfaces Firewall Services VPN Status Diagnostics Help

Firewall / Rules / LAN

The firewall rule configuration has been changed.
The changes must be applied for them to take effect. [Apply Changes](#)

Floating WAN LAN

Rules (Drag to Change Order)

☐	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☒	1 / 6.55 MIB	*	*	*	LAN Address	443 80	*	*		Anti-Lockout Rule	
☐	26 / 986.36 MIB	IPv4 *	LAN net	*	*	*	*	none		Default allow LAN to any rule	
☐	0 / 0 B	IPv6 *	LAN net	*	*	*	*	none		Default allow LAN IPv6 to any rule	

[Add](#) [Add](#) [Delete](#) [Save](#) [Separator](#)

12:30 AM 6/30/2021 17°C

31. The selected rules will be deleted.

pfSense.localdomain - Firewall: R x

Not secure | 10.10.1.1/firewall_rules.php?if=lan

System Interfaces Firewall Services VPN Status Diagnostics Help

Firewall / Rules / LAN

The changes have been applied successfully. The firewall rules are now reloading in the background.
[Monitor the filter reload progress.](#)

Floating WAN LAN

Rules (Drag to Change Order)

☐	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☒	1 / 6.57 MIB	*	*	*	LAN Address	443 80	*	*		Anti-Lockout Rule	
☐	26 / 986.36 MIB	IPv4 *	LAN net	*	*	*	*	none		Default allow LAN to any rule	
☐	0 / 0 B	IPv6 *	LAN net	*	*	*	*	none		Default allow LAN IPv6 to any rule	

12:30 AM 6/30/2021

32. To reboot pfSense, navigate to Diagnostics --> Reboot option from the main menu.

pfSense.localdomain - Firewall: R x +

Not secure | 10.10.1.1/firewall_rules.php?if=lan

pfSense COMMUNITY EDITION System Interfaces Firewall Services VPN Status Diagnostics Help

Firewall / Rules / LAN

The changes have been applied successfully. The firewall rules are now reloading in the background. Monitor the filter reload progress.

Floating WAN LAN

Rules (Drag to Change Order)

	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	...
☒	1 / 6.57 MIB	*	*	*	LAN Address	443	*	*	...
☐	26 / 986.36 MIB	IPv4 *	LAN net	*	*	*	*	none	...
☐	0 / 0 B	IPv6 *	LAN net	*	*	*	*	none	...

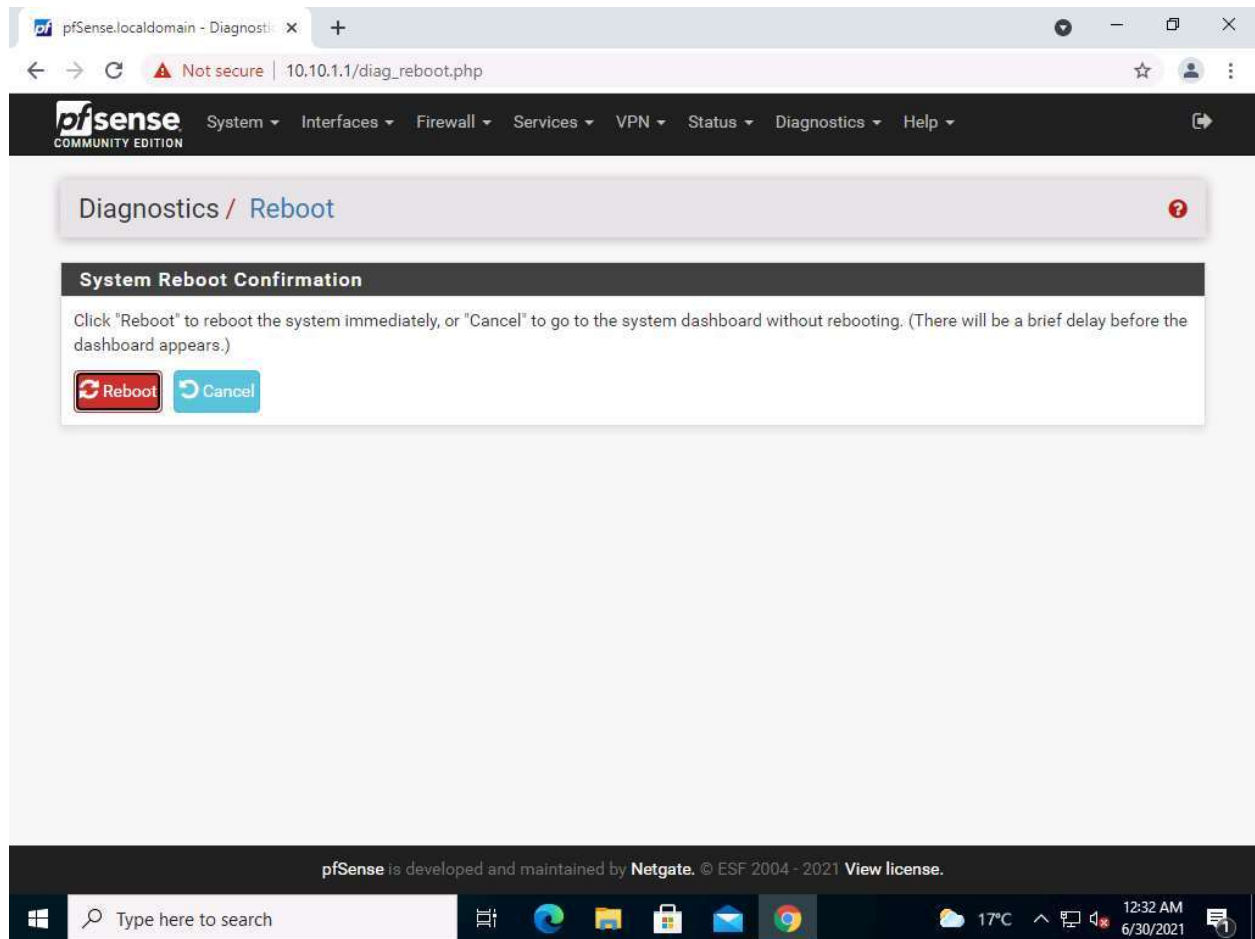
Reboot

10.10.1.1/diag_reboot.php

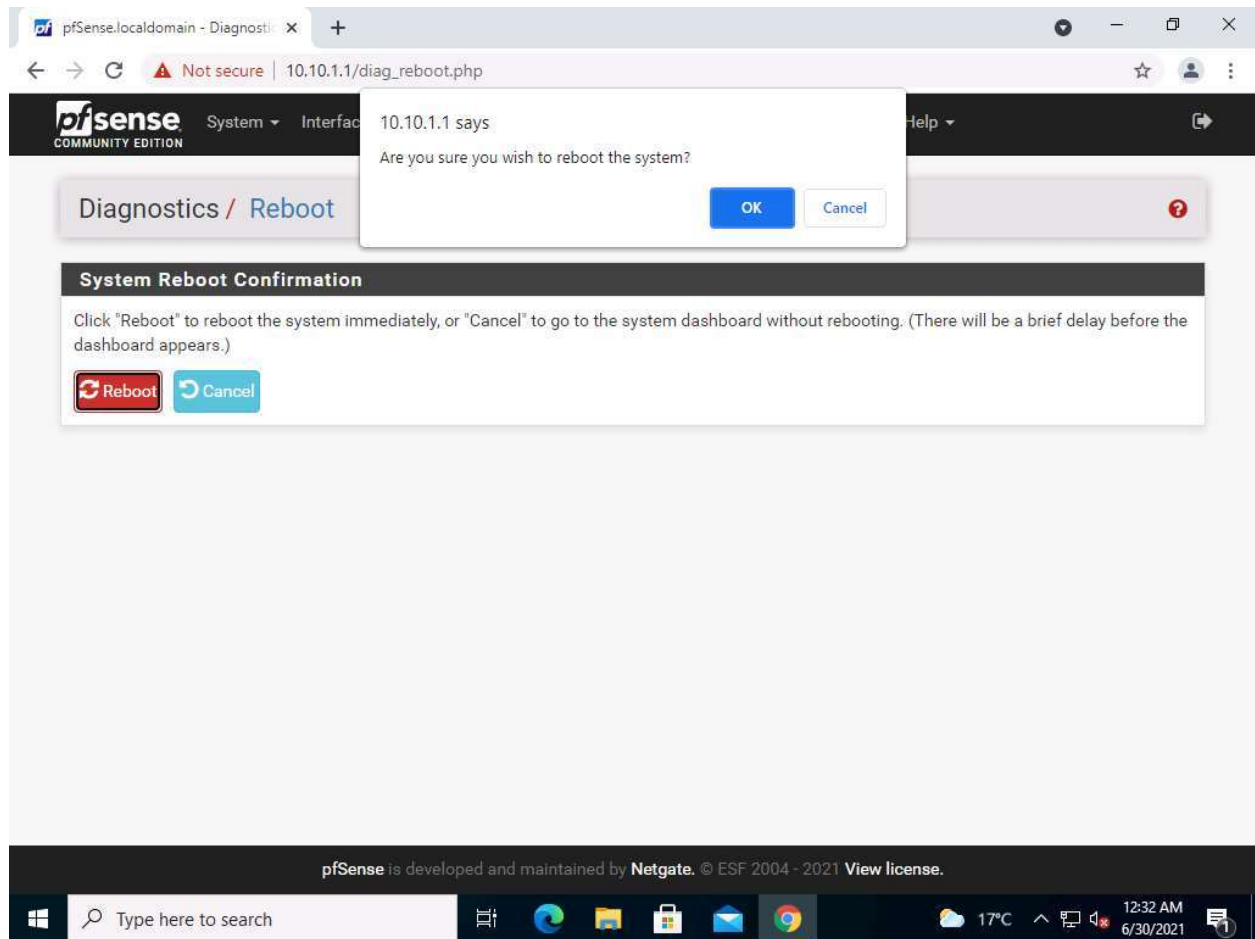
Type here to search

17°C 12:31 AM 6/30/2021

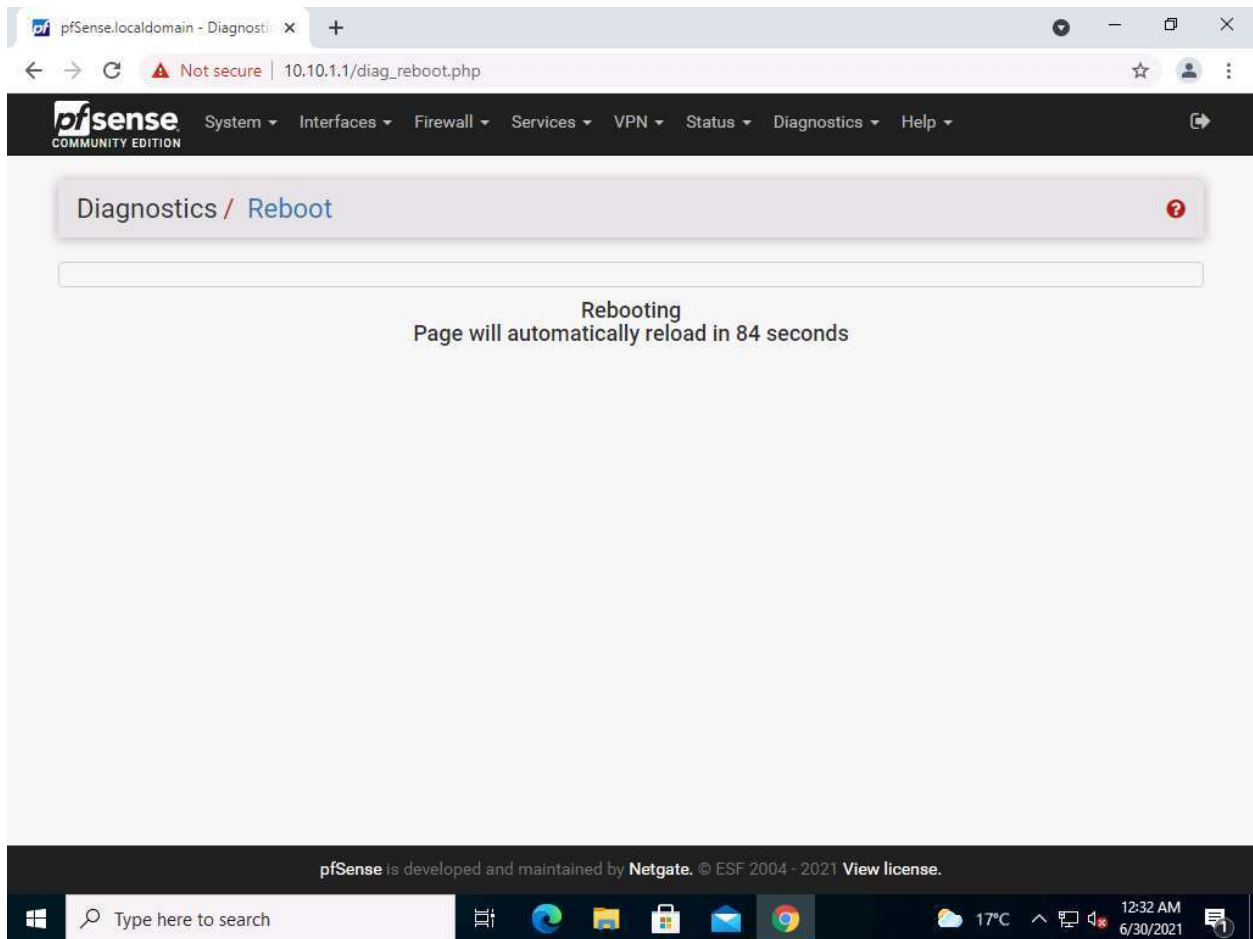
33. Click on Reboot button.



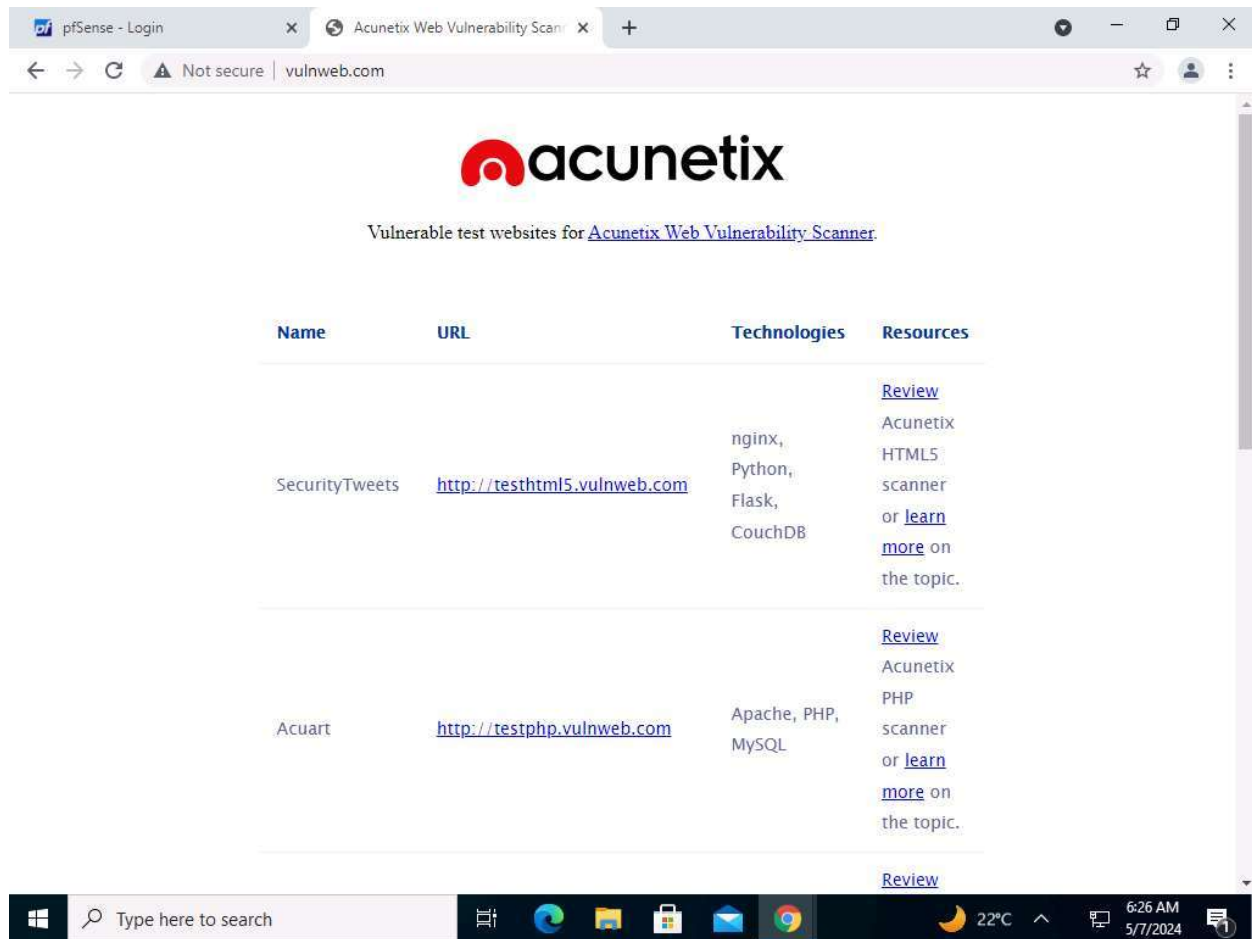
34. Click OK in the Reboot confirmation pop-up.



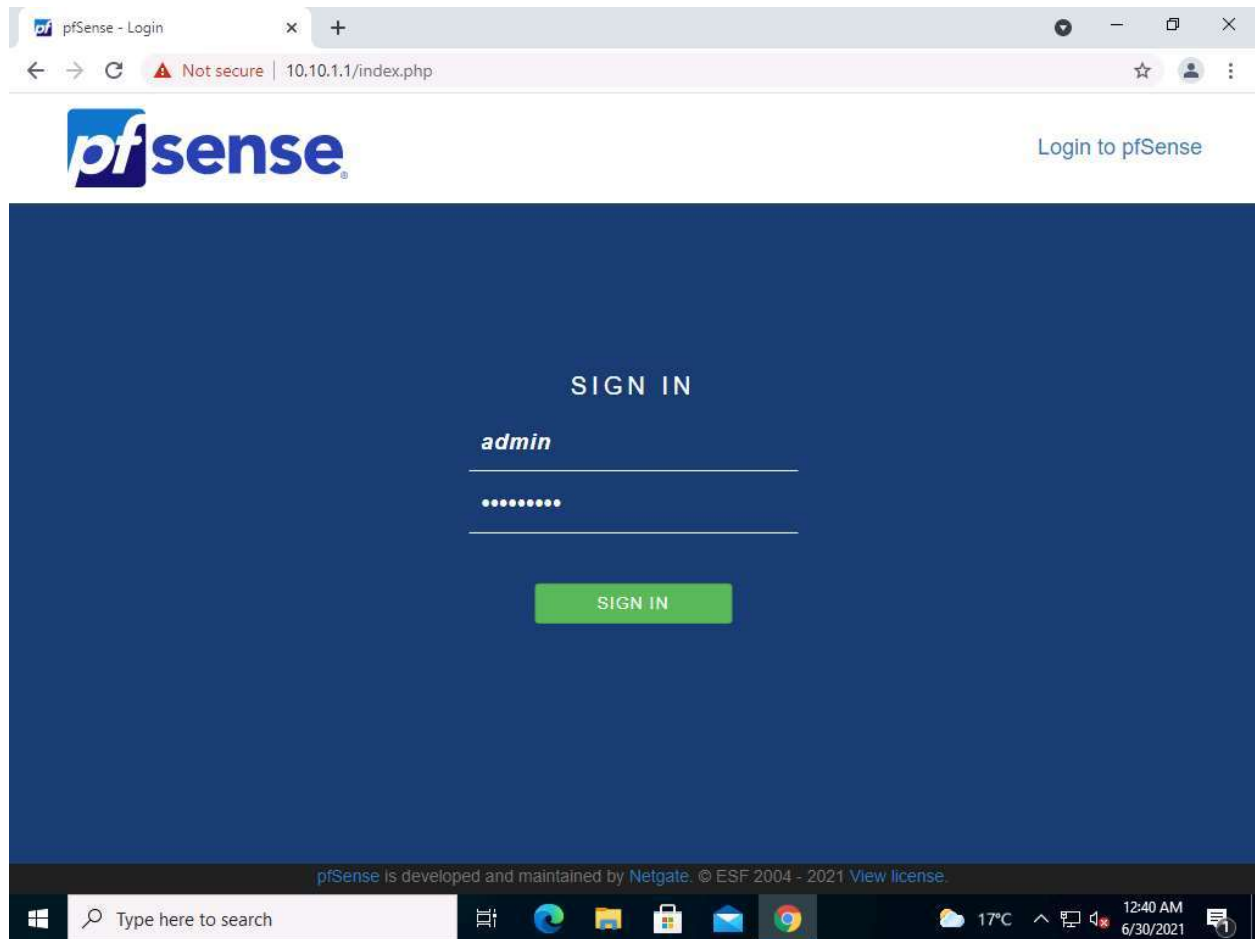
35. The pfSense firewall will reboot and load automatically.



36. The pfSense Sign in page will appear.
37. Now Open a new tab in Chrome browser and type `http://www.vulnweb.com` in the address bar, and press Enter. You will be able to access the web page as shown in the below screenshot:



38. Close the tab and navigate to the pfSense SIGN IN page.
39. Now we will configure schedules for time based rules
40. Type the username admin and password admin@123, and click the SIGN IN button as shown in the screenshot below.



41. The pfSense Dashboard will appear. Navigate to Firewall-->Schedules from the main menu.

The screenshot shows the pfSense web interface in a browser window. The browser's address bar shows 'pfSense.localdomain - Status: D...' and the URL 'https://10.10.1.1/firewall_schedule.php'. The pfSense header includes the logo and navigation tabs: System, Interfaces, Firewall (selected), Services, VPN, Status, Diagnostics, and Help. A dropdown menu for the Firewall tab is open, showing options: Aliases, NAT, Rules, Schedules (highlighted with a red box), Traffic Shaper, and Virtual IPs. The main content area is divided into two panels. The left panel, titled 'Status / Dashboard', contains a 'System Information' table. The right panel, titled 'Netgate Services And Support', displays contract information and support resources.

System Information	
Name	pfSense.localdomain
User	admin@10.10.1.2 (Local D
System	Microsoft Azure Netgate Device ID: 9419252c0bb6c3ee4f14
BIOS	Vendor: American Megatrends Inc. Version: 090006 Release Date: Thu Apr 28 2016
Version	2.4.5-RELEASE-p1 (amd64) built on Tue Jun 02 17:51:17 EDT 2020 FreeBSD 11.3-STABLE Version 2.5.1 is available. Cloud icon Version information updated at Wed Jun 30 4:33:55 UTC 2021 Refresh icon
CPU Type	Intel(R) Xeon(R) Gold 6262V CPU @ 1.90GHz AES-NI CPU Crypto: Yes (inactive)
Kernel PTI	Disabled
MDS Mitigation	Inactive

https://10.10.1.1/firewall_schedule.php 18 Minutes 11 Seconds

Netgate Services And Support

Contract type: Community Support
Community Support Only

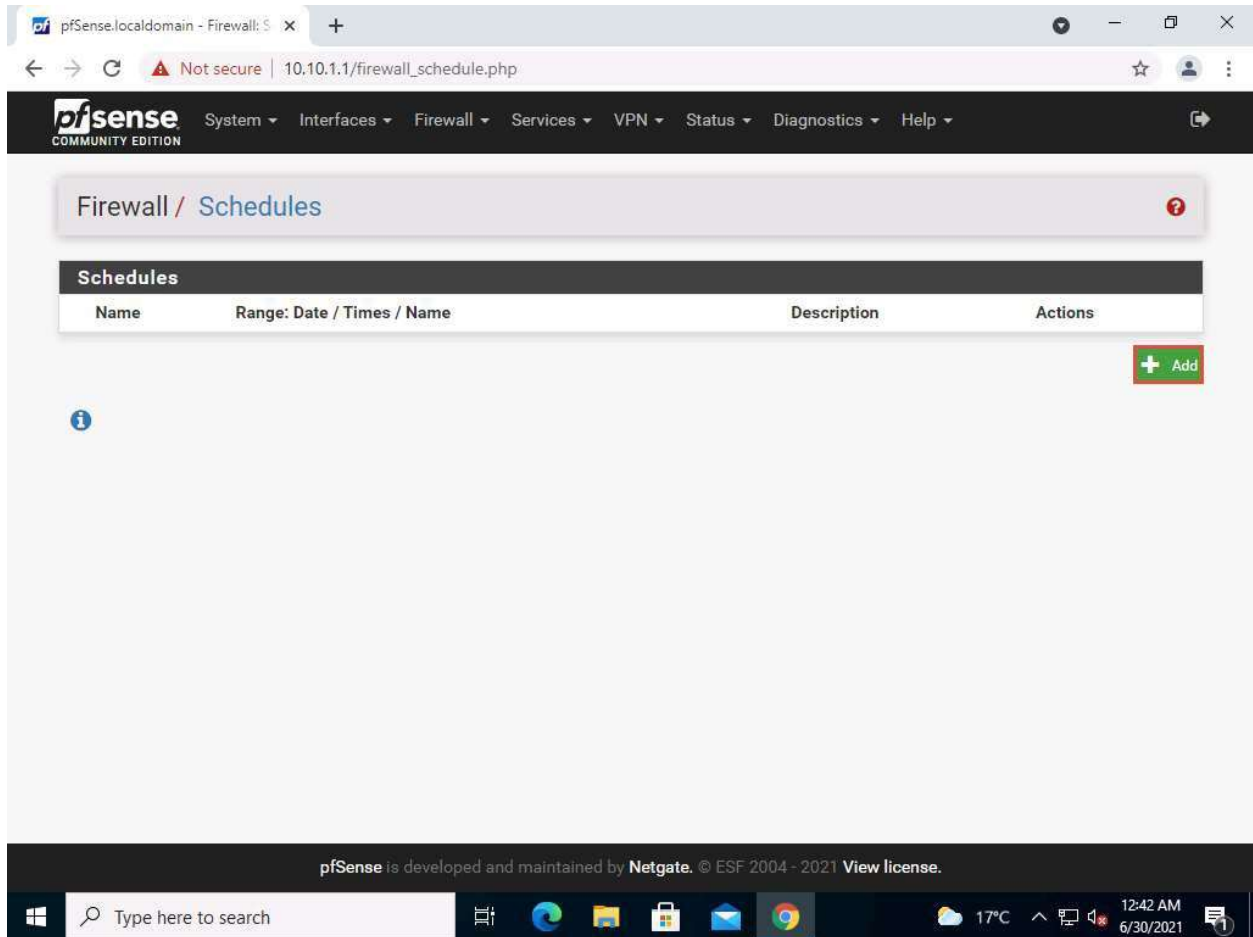
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42. Click + Add button in the Firewall/ Schedules window.



43. The Firewall/ Schedules/ Edit window opens. Enter a name in the Schedule Name section under Schedule Information and enter description in the Description field .

Schedule name must only contain letters and digits, no spaces. Here we have given the name as WorkingHours and Normal Working Hours as Description.

pfSense.localdomain - Firewall: S x +

Not secure | 10.10.1.1/firewall_schedule_edit.php

System Interfaces Firewall Services VPN Status Diagnostics Help

Firewall / Schedules / Edit

Schedule Information

Schedule Name WorkingHours
The name of the schedule may only consist of the characters "a-z, A-Z, 0-9 and _".

Description Normal Working Hours
A description may be entered here for administrative reference (not parsed).

Month June_21

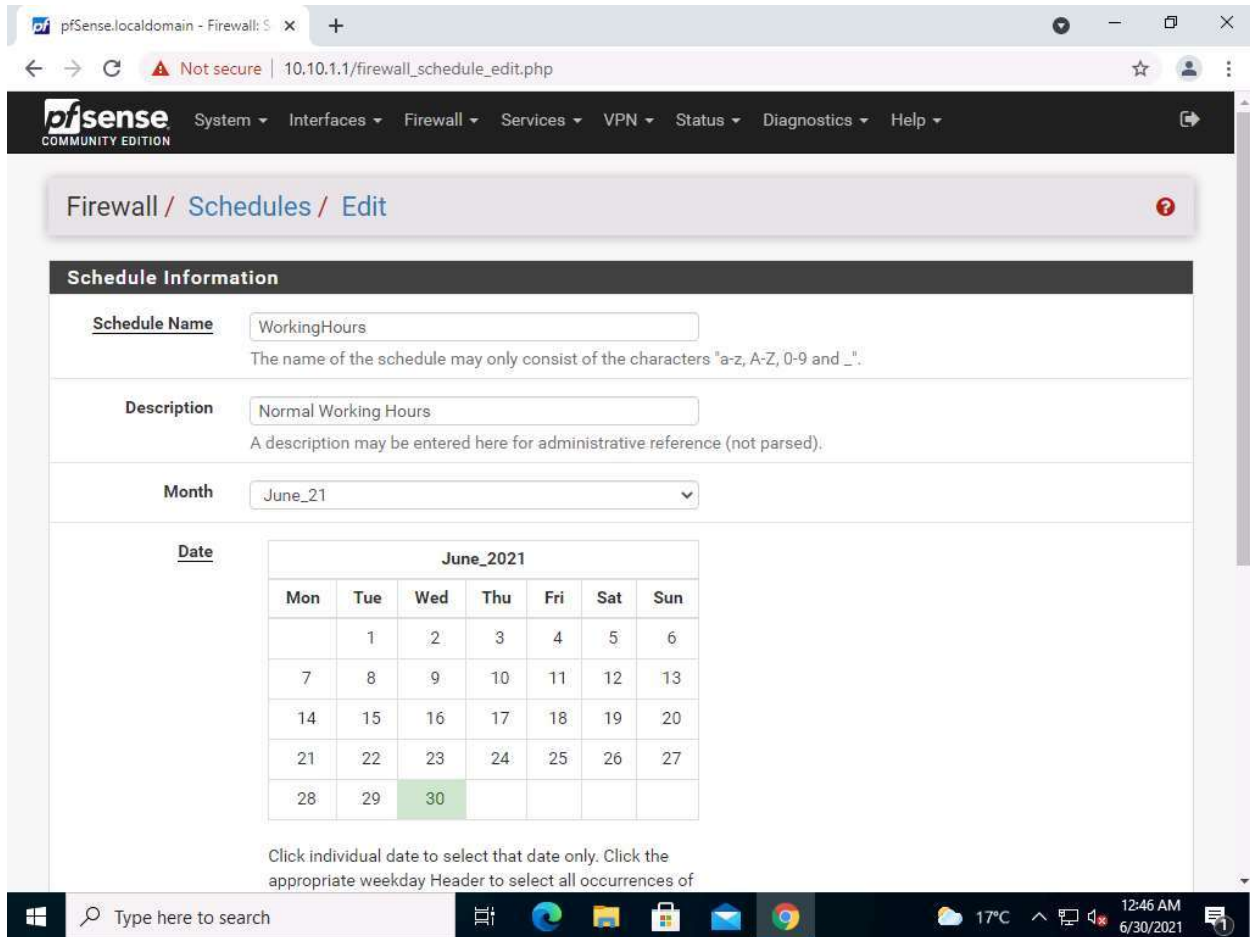
Date

June_2021						
Mon	Tue	Wed	Thu	Fri	Sat	Sun
	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30				

Click individual date to select that date only. Click the appropriate weekday Header to select all occurrences of

44. Now set the Month by selecting a specific month and days, or by clicking the day of the week header for weekly recurring schedules.

Here we are selecting a single day in a month.



45. Choose a Start Time and Stop Time which control when the rule is active on the selected days. Enter an optional Time Range Description for this specific range.

The time cannot cross midnight on any day. A full day is 0:00 to 23:59.

While performing the task in the lab, make sure that the time you are selecting is 5 to 10 minutes ahead of the time.

pfSense.localdomain - Firewall: S x +

Not secure | 10.10.1.1/firewall_schedule_edit.php?id=0

Mon	Tue	Wed	Thu	Fri	Sat	Sun
	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30				

Click individual date to select that date only. Click the appropriate weekday Header to select all occurrences of that weekday.

Time

Start Hrs: 1 Start Mins: 15 Stop Hrs: 23 Stop Mins: 59

Select the time range for the day(s) selected on the Month(s) above. A full day is 0:00-23:59.

Time range description

Work day

A description may be entered here for administrative reference (not parsed).

+ Add Time Clear selection

Configured Ranges

Save

Type here to search

17°C 1:24 AM 6/30/2021

46. Click on + Add Time to add the choices as range, and click Save.

pfSense.localdomain - Firewall: S

+

Not secure | 10.10.1.1/firewall_schedule_edit.php?id=0

☆

⋮

7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30				

Click individual date to select that date only. Click the appropriate weekday Header to select all occurrences of that weekday.

Time

1

15

23

59

Start Hrs

Start Mins

Stop Hrs

Stop Mins

Select the time range for the day(s) selected on the Month(s) above. A full day is 0:00-23:59.

Time range description

Work day

A description may be entered here for administrative reference (not parsed).

+ Add Time

Clear selection

Configured Ranges

Save

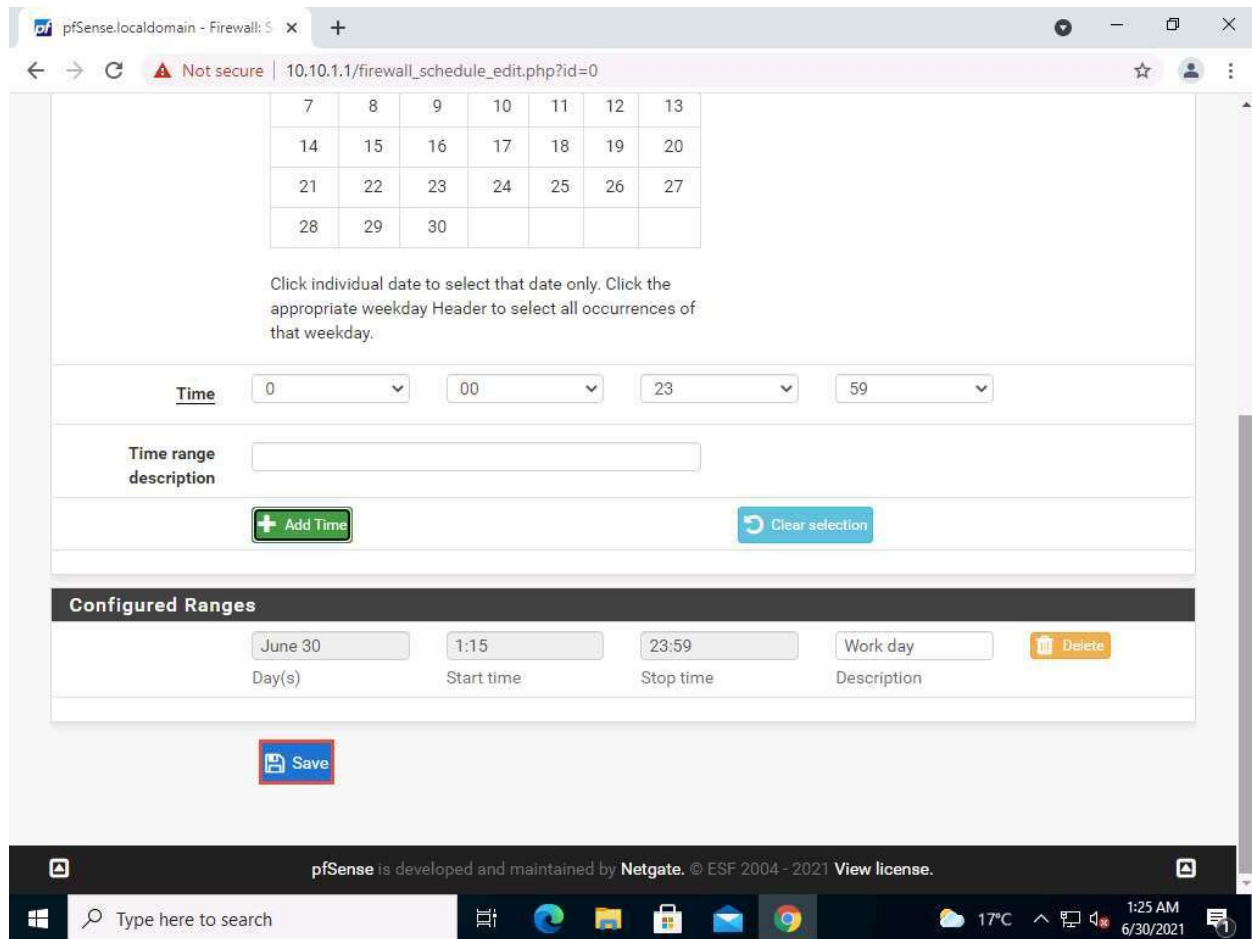
pfSense is developed and maintained by Netgate. © ESF 2004 - 2021 View license.

Windows Taskbar

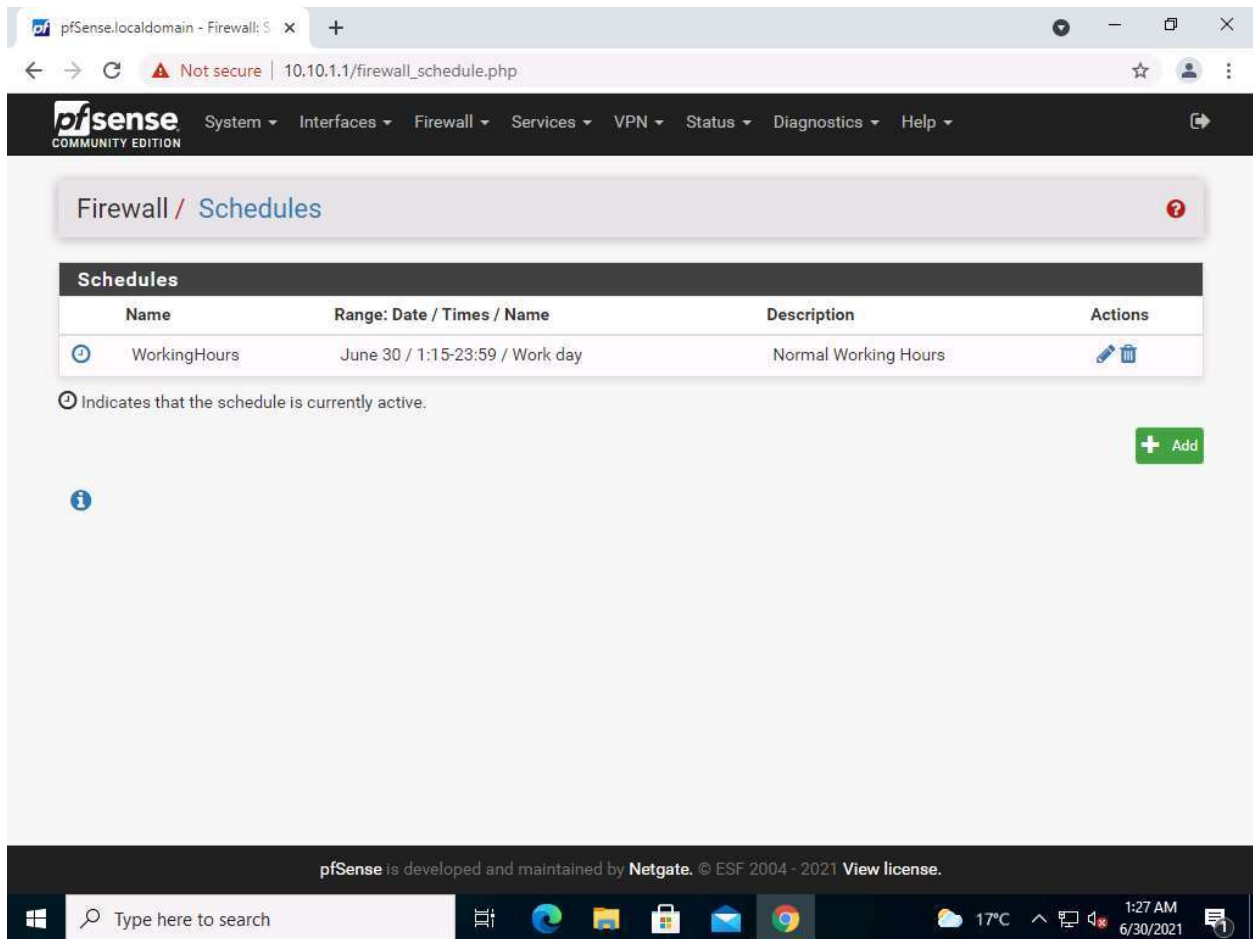
Type here to search

17°C

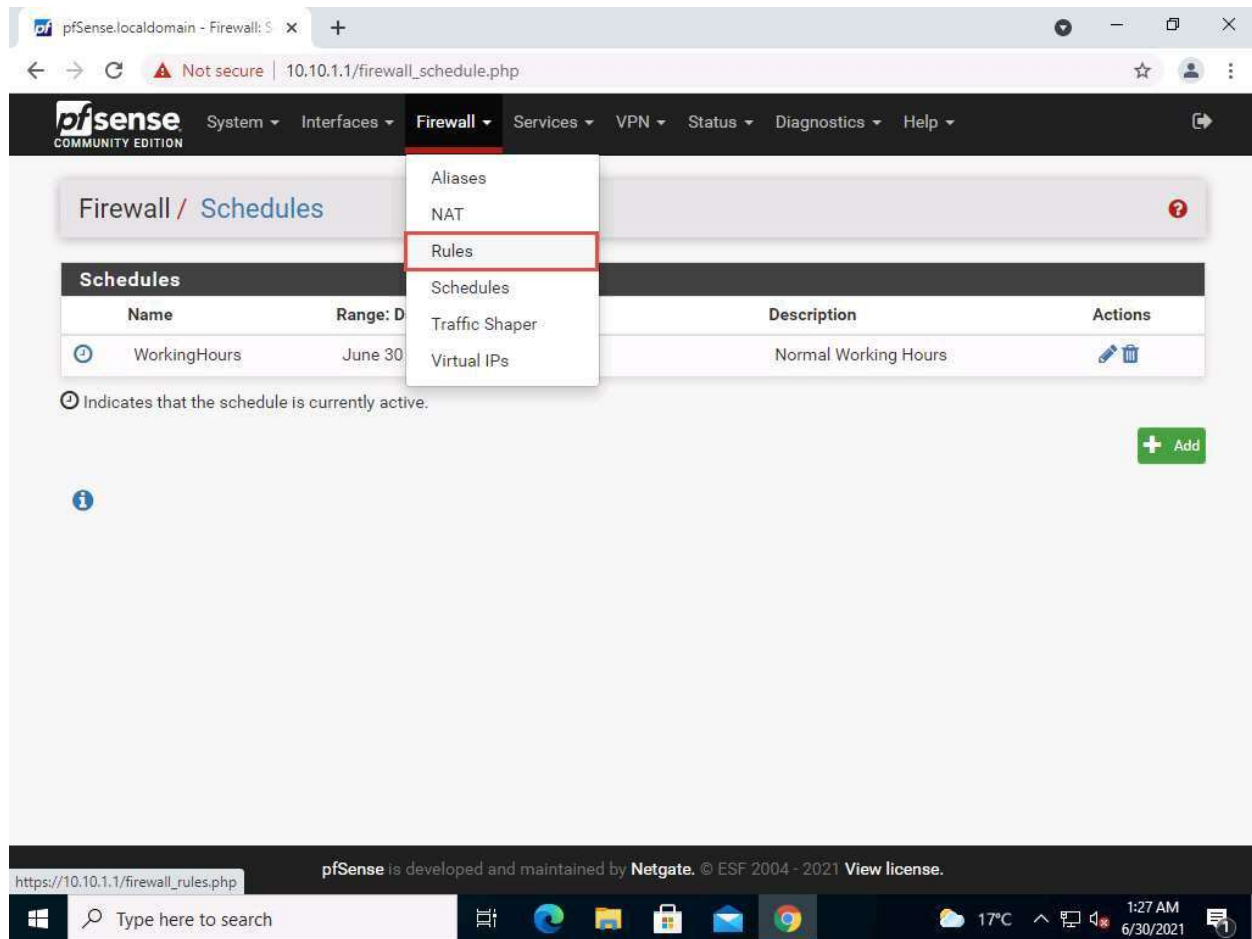
1:24 AM 6/30/2021



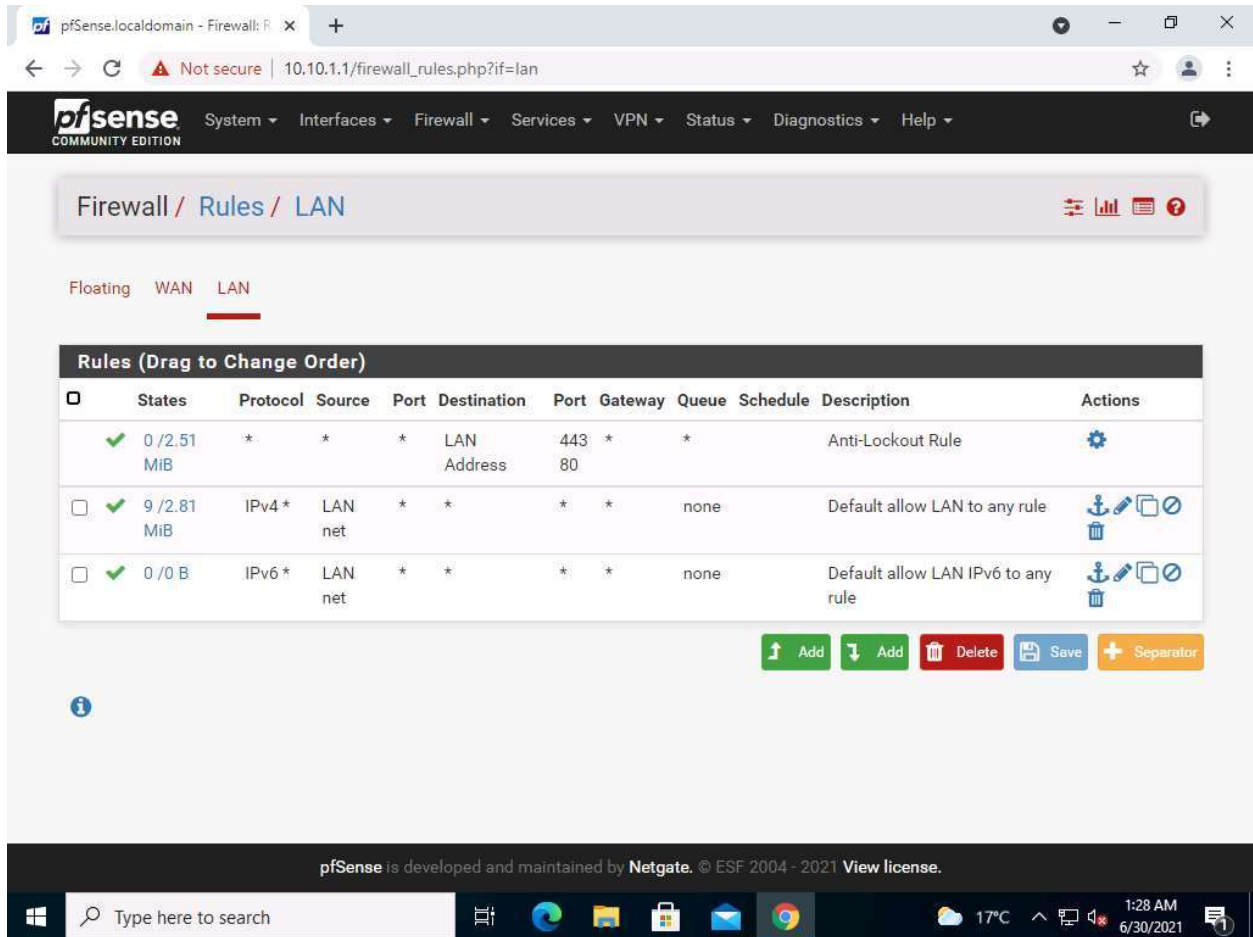
47. You will be redirected to the Firewall/ Schedules page with the new schedule listed.



48. To create a new firewall rule navigate to Firewall-->Rules from the main menu.



49. The Firewall/Rules/WAN page will appear. Click the LAN option.



50. To create a rule, click the up arrow Add button.

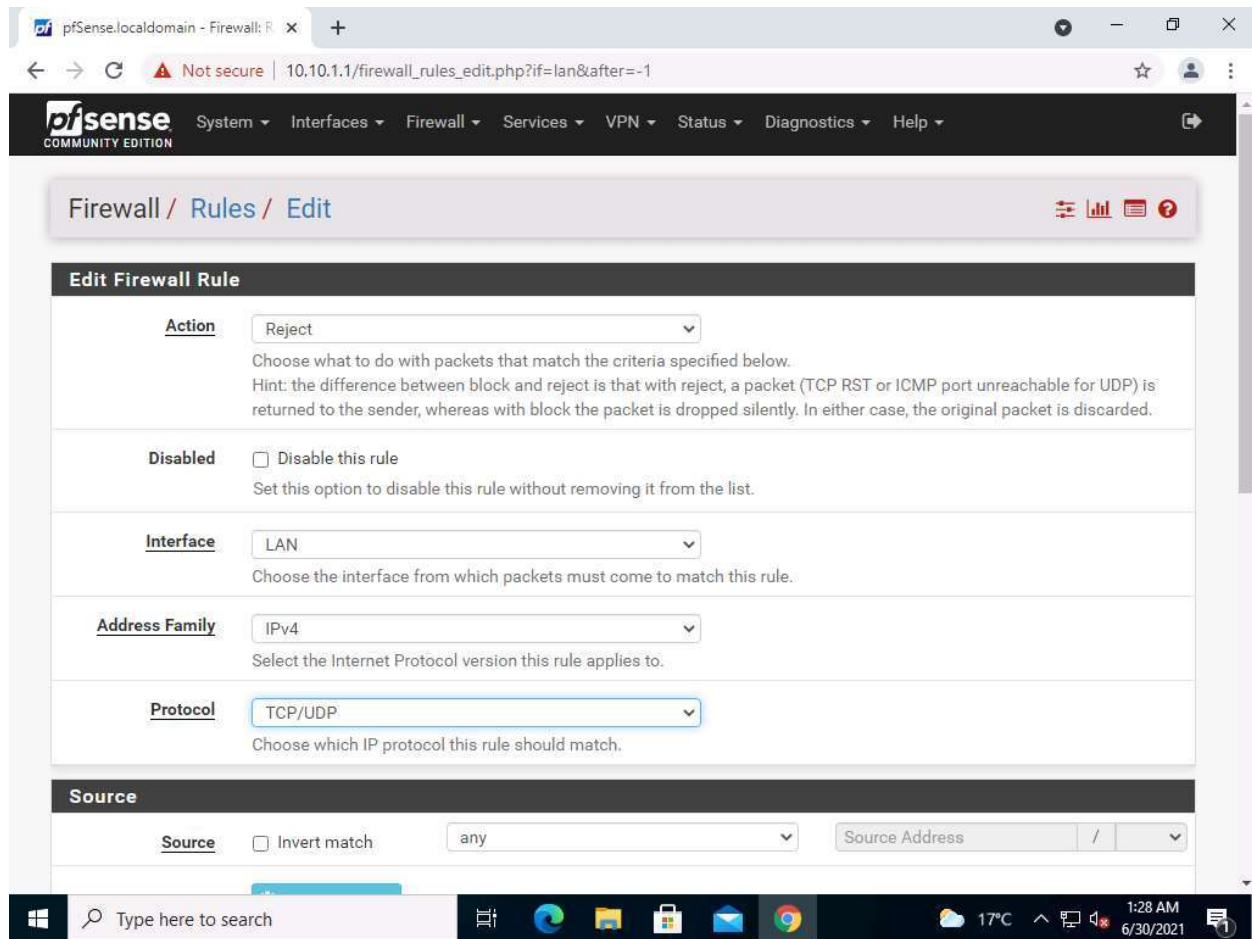
51. Set following details under Edit Firewall Rule section

Action --> Reject

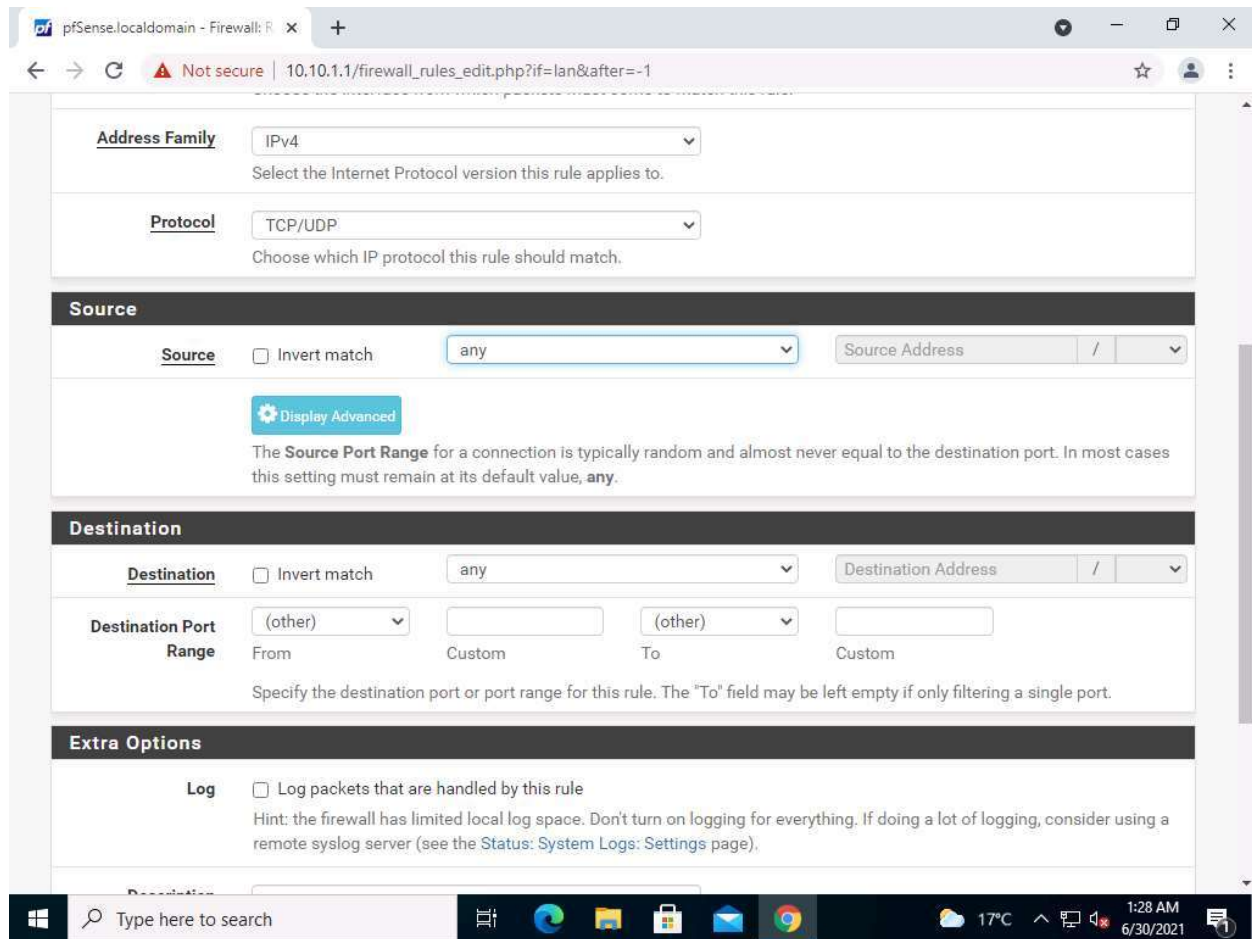
Interface --> LAN

Address Family --> IPv4

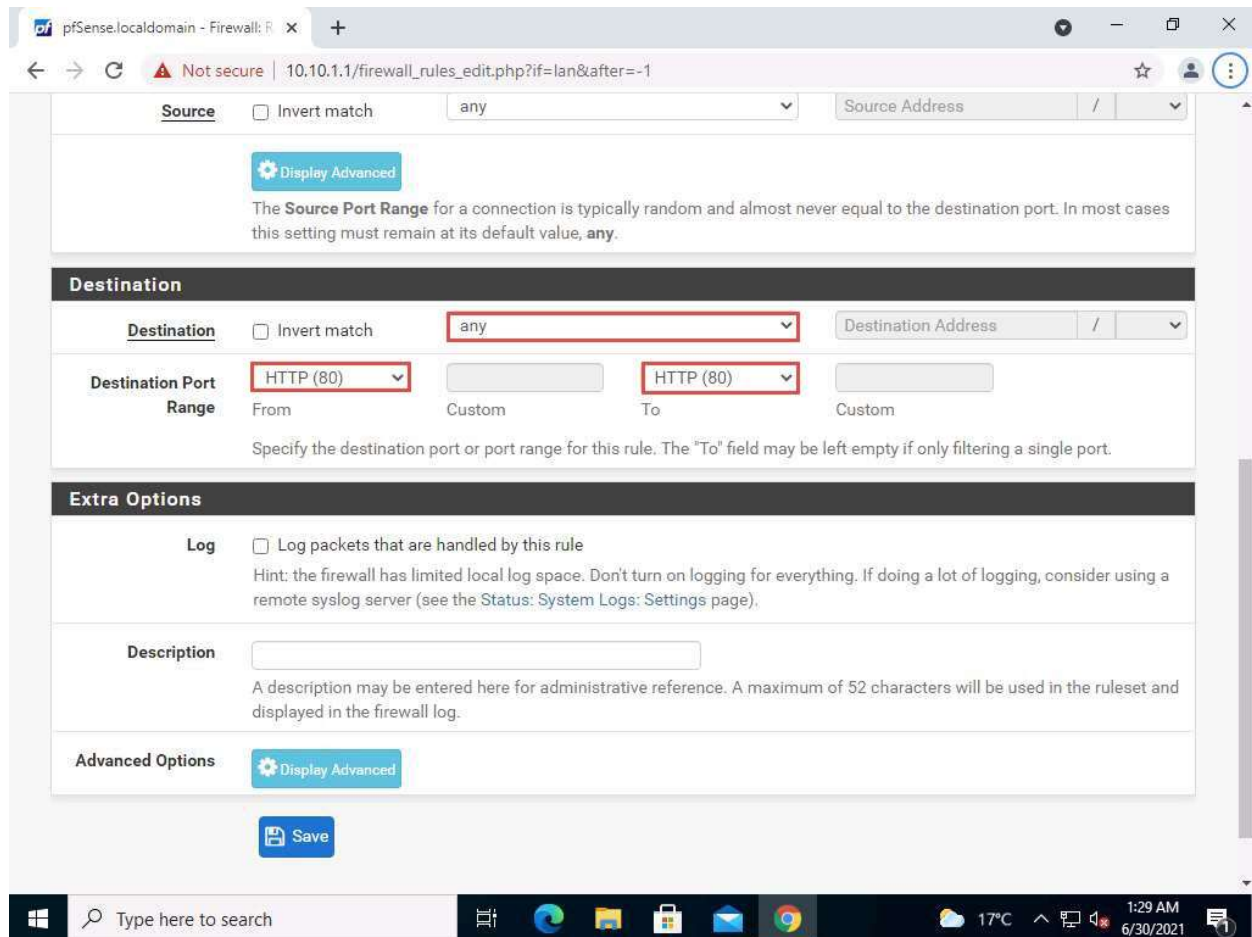
Protocol --> TCP/UDP.



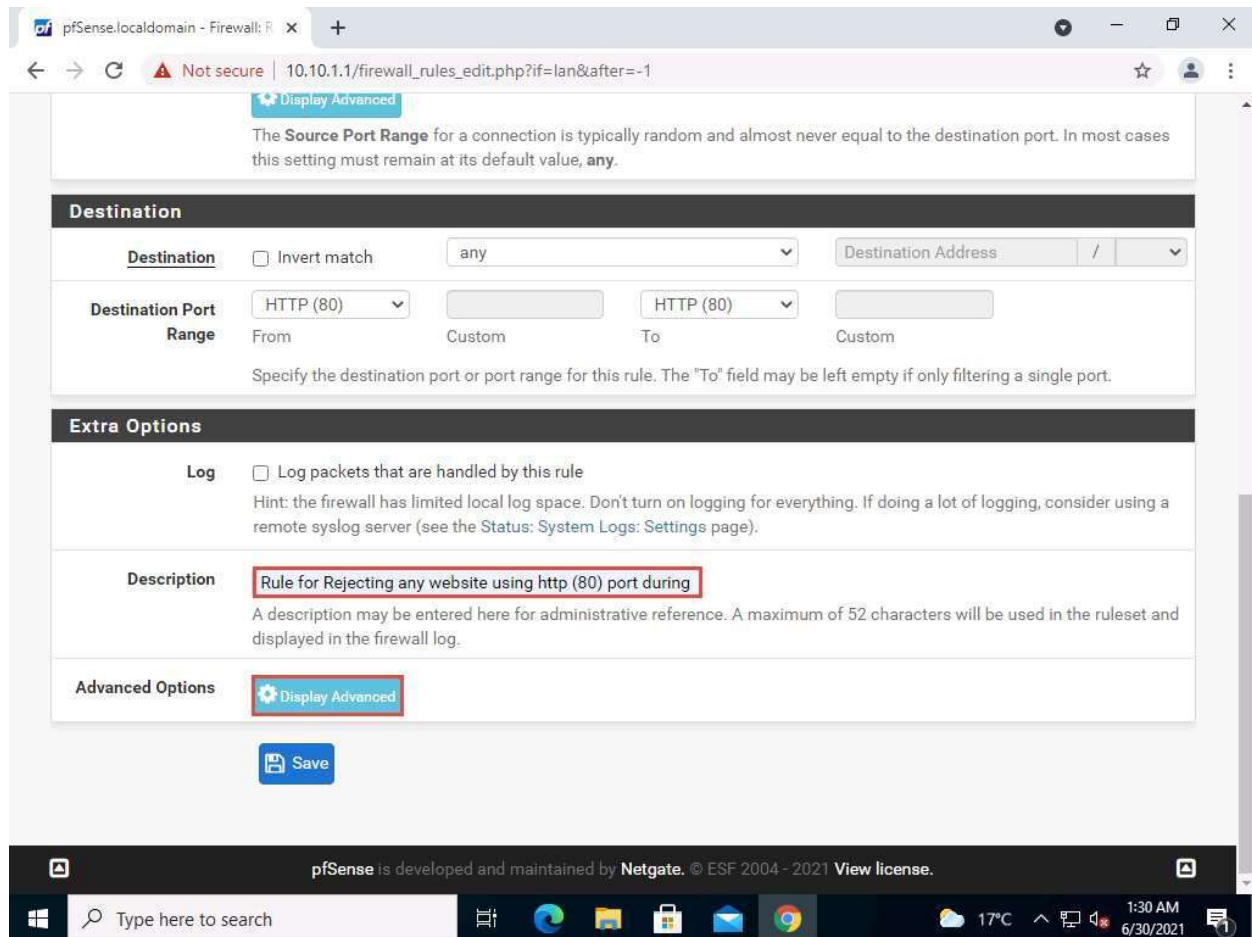
52. Under Source section, select any from the dropdown as shown in the below screenshot.



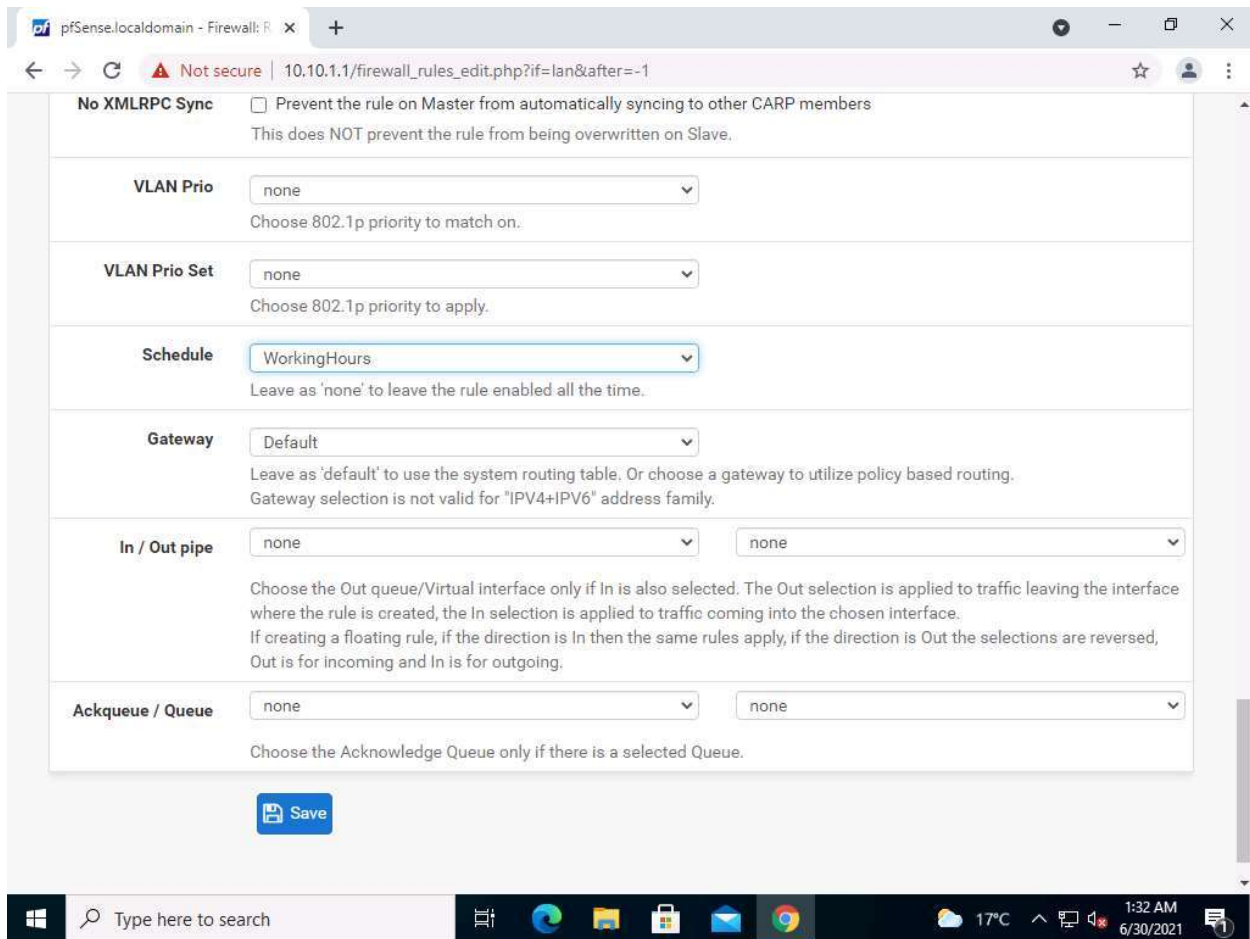
53. Under Destination section, select any from the dropdown and set Destination Port Range to HTTP (80) from the dropdown.



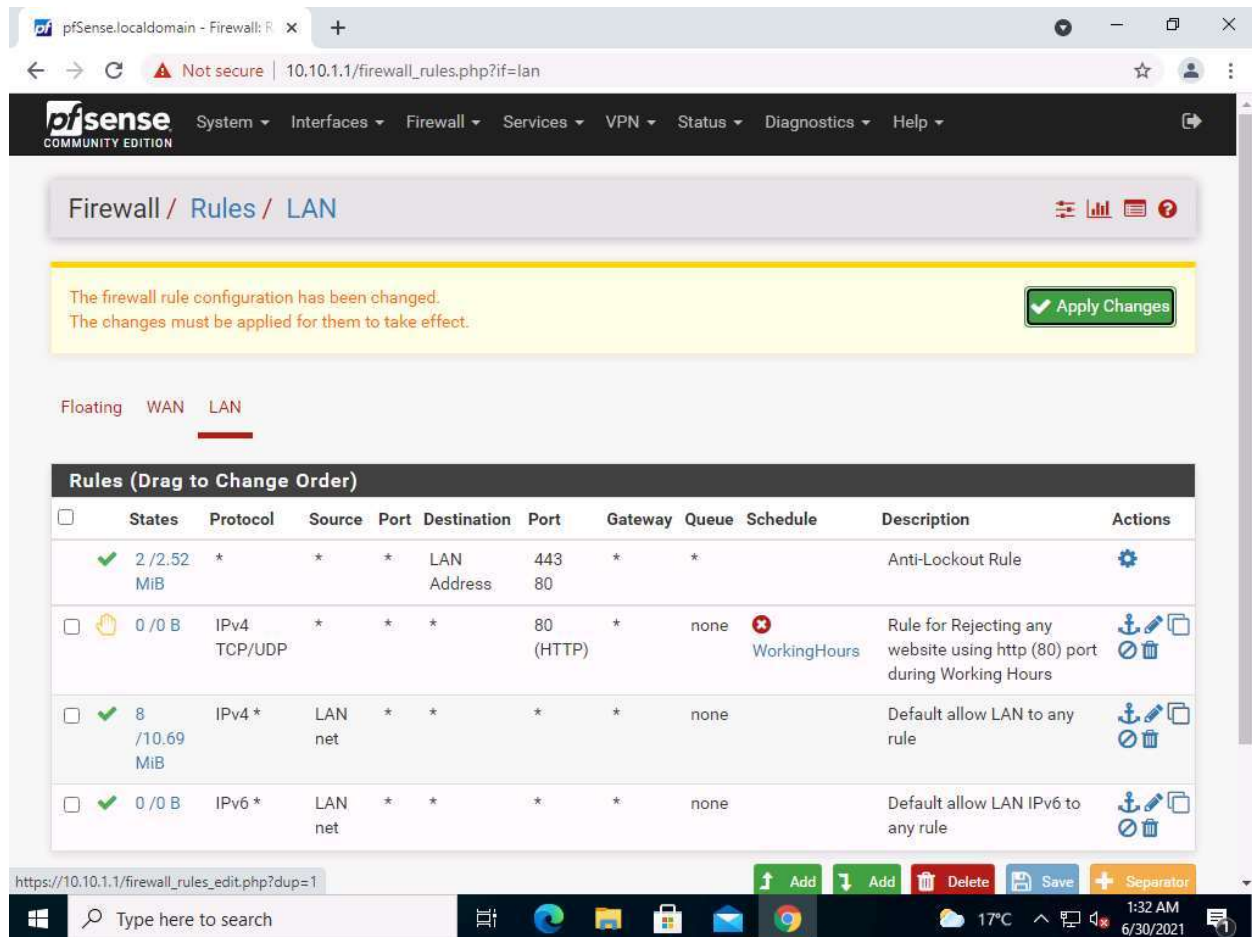
54. Scroll down. Under Extra Options, enter Rule for Rejecting any website using http (80) port during Working Hours in the Description field, and click on Display Advanced.



55. In Advanced Options go to Schedule section and select the newly created WorkingHours Schedule from the drop down.



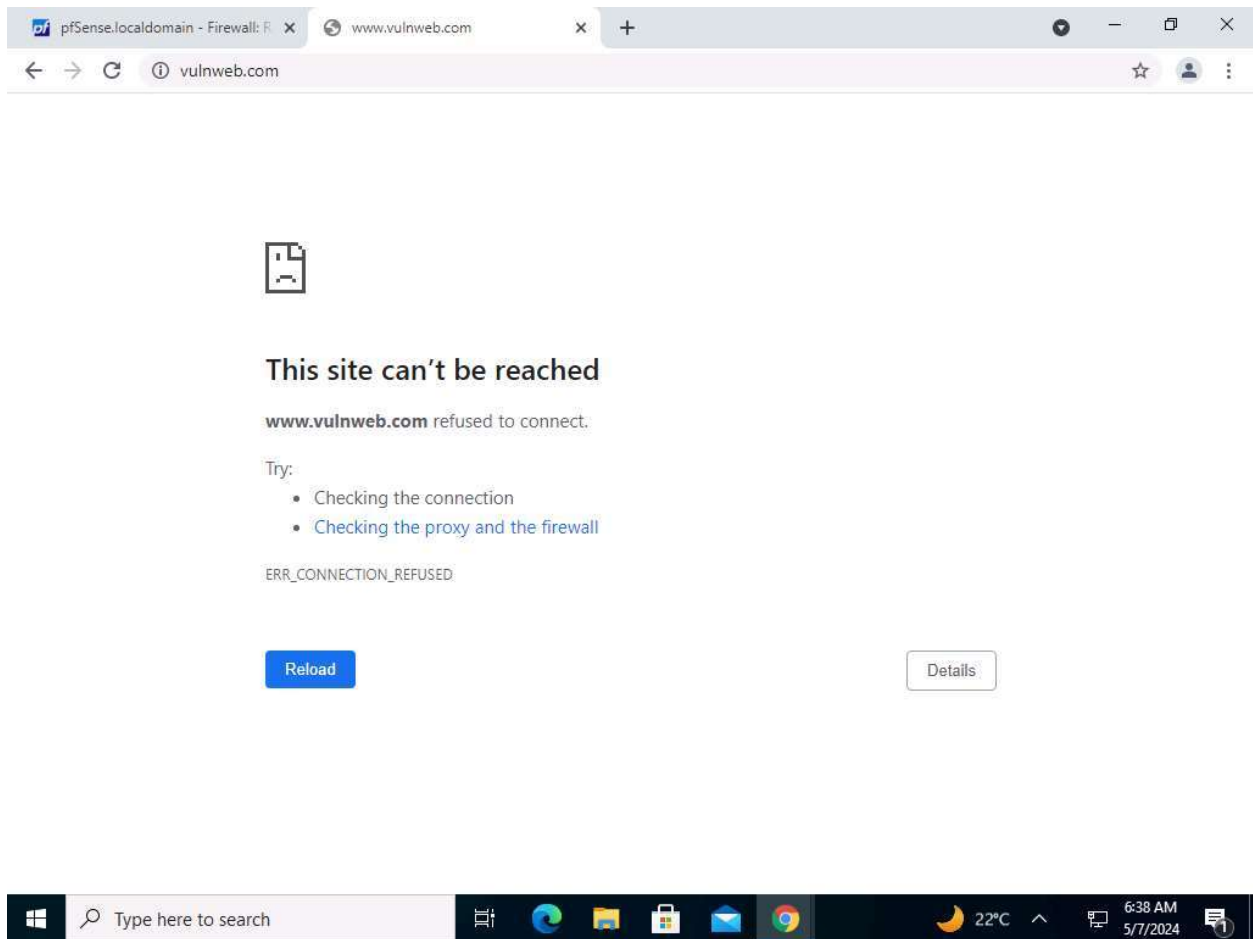
56. Leaving the other options set to default scroll down the page and click on Save.
57. The page will redirect to the Firewall/Rules/LAN page. Click Apply Changes.



58. Reload the webpage and open a new tab in chrome browser and type <http://vulnweb.com/> in the address bar, and press Enter

If a pop-up appears while reloading webpage, click on Continue.

If changes are not affecting, then reboot the pfSense firewall and clear the Chrome browser cache.



59. You will see the message as This site can't be reached. This is because the pfSense firewall rule is now preventing port http.